

Private Pilot

Flight Lesson Guide

An original instructor flight guide aligned to the current publisher lesson order, factual preparation assignments and individual lesson-page time targets. The source is an Academy Part 141 outline; this guide uses Part 61 progression and grants no school graduation or examining authority. Parked discussion/preflight are separate from the timed complete-flight sequence. Early lessons permit appropriate assistance; solo and assessment gates require demonstrated independent safety. Broad source lists that exceed a useful practice budget carry named unfinished work to extended or additional flights.

Sporty's Academy Private Pilot TCO V20, revised June 17, 2026; original Part 61 teaching adaptation

Use this guide

Read the full teaching guide before flight. Use the compact checklist to sequence the flight. Minutes are pacing targets; repeat, stop, or defer an exercise when proficiency, conditions, or available time require it. Checkmarks indicate items covered, not a proficiency endorsement.

Aircraft values come from the applicable POH/AFM and approved checklist. Brief the aircraft-specific speeds, configurations, limitations, and any simulation method before conducting a maneuver.

Entry and preparation

- Work with an authorized CFI in the actual aircraft. Establish student certificate/medical eligibility and the training/endorsement conditions applicable to solo before release.
- Use the aircraft POH/AFM, installed supplements, current FAA standards and actual weather/airport data to set every operating value.
- Use Part 61 requirements to reconcile actual experience; publisher lesson targets do not automatically satisfy total, dual, solo, cross-country, night, instrument or recent practical-test preparation requirements.

Sources and alignment

Sporty's Academy Private Pilot Training Course Outline V20 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Private Pilot Airplane ACS - FAA-S-ACS-6C, effective May 31, 2024

Part 61 student solo and airplane experience rules - Current eCFR, verified October 10, 2026

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Control and solo readiness

Ground preparation, basic control, low-speed/wind/emergency skills and independent pattern evidence before a conditional first solo.

Lesson 1 · Aircraft preparation

Study airworthiness, documents, inspections and a deliberate walkaround before the first flight.

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Ground preparation objectives

- Study airworthiness, documents, inspections and a deliberate walkaround before the first flight.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 9](#)

[Sporty's course preparation: Vol 1: Segments 1-7, 12, 13, 26](#)

[Sporty's course preparation: Vol 3: Segment 16](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 2 · First flight: control, trim and the traffic pattern

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72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-32 Air 08-24	[] Sight picture, climb, level flight and descent	1.
32-42 Air 24-34	[] Airspeed changes and trim	1.
42-58 Air 34-50	[] Coordinated turns to headings	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 2

Introduce the airplane as a controllable system. Build an outside sight picture and positive control transfer; the instructor still demonstrates and assists takeoff and landing.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Introduce the airplane as a controllable system. Build an outside sight picture and positive control transfer; the instructor still demonstrates and assists takeoff and landing.
- Use outside references for elementary pitch, bank and coordination; assistance is expected.
- Recognize the complete preflight-to-secured-aircraft workflow without claiming independent landing proficiency.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Sight picture, climb, level flight and descent; Airspeed changes and trim; Coordinated turns to headings.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Model each action and explain it before the learner attempts it. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Demonstrate the outside references and coach control inputs; retain responsibility for the landing/takeoff outcome. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Sight picture, climb, level flight and descent · 16 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: At a safe cruise altitude, point out the horizon position for level flight. Show how a small pitch change moves the horizon, then coordinate power for a climb. Anticipate level-off, set cruise power and trim. Show a descent and level-off in the same order.

Student exercise: 1. Describe the outside attitude and cross-check the instruments. 2. Hold straight flight with coordinated rudder. 3. Establish the briefed climb airspeed and level at the assigned altitude. 4. Establish a descent, then restore level flight. 5. State what pitch and power changed.

Practice: One instructor sequence, then two student climb-level-descent sequences; use smaller changes if the learner is overloaded.

Look for: Recognizes the changing sight picture, coordinates pitch and power and levels with timely coaching; initial-flight assistance is expected.

Watch / correct: Chasing the vertical-speed indication - return eyes to the horizon and make one small pitch correction.

Watch / correct: Ball displaced during a power change - add the appropriate rudder while keeping wings level.

Stop / transition: Restore ordinary cruise if traffic, turbulence or workload prevents an outside scan; instructor takes control before an unsafe altitude or airspeed develops.

Reference: AFH: basic flight maneuvers; PHAK: flight controls and instruments

Airspeed changes and trim · 10 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show cruise-to-low-cruise airspeed by setting power, holding altitude with pitch and trimming only after the attitude is established. Reverse the sequence to cruise and demonstrate how trim relieves force without setting the flight path.

Student exercise: 1. Choose the assigned airspeed and configuration within the POH limits. 2. Change power while holding altitude. 3. Stabilize the attitude, then trim. 4. Return to cruise. 5. Explain which control establishes attitude and which relieves pressure.

Practice: Two complete speed transitions, with one hands-light trim check after stabilization.

Look for: Controls altitude through a speed change, avoids chasing trim and verifies the airplane stays stable when control pressure is relaxed.

Watch / correct: Using trim as the primary pitch control - establish attitude with the elevator before trimming.

Watch / correct: Adding power without compensating pitch - anticipate the attitude change and recheck altitude.

Stop / transition: Leave the exercise before stall warning, a configuration limit or an unsafe altitude trend; use the recovery already taught rather than continuing a speed target.

Reference: AFH: straight-and-level flight, transitions and trim

Coordinated turns to headings · 16 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show clearing, an outside bank reference and coordinated rudder. Compare a shallow turn with a medium-bank turn, then show leading the rollout to capture a heading without skidding.

Student exercise: 1. Clear toward the turn. 2. Enter the briefed bank with coordinated controls. 3. Maintain the outside pitch picture. 4. Anticipate rollout to the chosen heading. 5. Re-trim only when stable and reverse direction after another clearing scan.

Practice: Two turns in each direction, using different heading changes and one deliberate rollout correction.

Look for: Maintains a coordinated turn, identifies altitude drift and rolls out smoothly with decreasing instructor assistance.

Watch / correct: Staring at the heading instrument - use it briefly and return to the outside reference.

Watch / correct: Using rudder alone to hurry the rollout - reduce bank with coordinated aileron and rudder.

Stop / transition: Roll wings level and discontinue if the turn is uncoordinated, traffic appears or the learner loses the altitude/airspeed picture.

Reference: AFH: turns; PHAK: adverse yaw and coordination

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Demonstrate the pattern, aiming picture, flare and directional control, then allow only the participation the learner can safely manage. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.

- Student can follow the preparation/start/taxi workflow and identify basic flight attitudes with assistance; independent takeoff or landing mastery is not expected.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 3, 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 6 & 8

Sporty's course preparation: Vol 1: Segments 14, 19, 21-24

Sporty's course preparation: Vol 2: Segments 2, 11 -13

Sporty's course preparation: Vol 3: Segments 4, 15

Sporty's course preparation: Vol 5: Segment 8

Sporty's course preparation: Vol 6: Segment 6

Sporty's Academy Private TCO V20: lesson 2 - source page 21 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 3 · Airport and taxi preparation

Prepare airport signs, markings, pattern rules and runway-incursion prevention before further taxi/pattern practice.

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Ground preparation objectives

- Prepare airport signs, markings, pattern rules and runway-incursion prevention before further taxi/pattern practice.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Airplane Flying Handbook: Chapters 2 & 8

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 14

Current Part 61 and applicable Part 91 rules

Current AIM: Aeronautical Information Manual - Chapter 2

Sporty's course preparation: Vol 1: Segments 15-18

Sporty's course preparation: Vol 2: Segment 14

Sporty's course preparation: Vol 3: Segment 24, 25

Sporty's course preparation: Vol 5: Segment 23

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 4 · Aerodynamics preparation

Prepare lift, angle of attack, drag and stall recognition for the first low-speed lesson.

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Ground preparation objectives

- Prepare lift, angle of attack, drag and stall recognition for the first low-speed lesson.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 5](#)

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 4 & 5](#)

[Sporty's course preparation: Vol 1: Segment 25](#)

[Sporty's course preparation: Vol 2: Segments 5-6, 9, 10](#)

[Sporty's course preparation: Vol 3: Segments 19-21](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 5 · Slow flight and first-indication stall recovery

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72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-34 Air 08-26	[] Slow flight: configuration, coordination and exit	1.
34-58 Air 26-50	[] Recognize and recover at the first stall indication	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 5

Introduce low-speed handling and recovery at the first stall cue. Keep the first stall lesson distinct from the later full-stall lessons.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Introduce low-speed handling and recovery at the first stall cue. Keep the first stall lesson distinct from the later full-stall lessons.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Slow flight: configuration, coordination and exit; Recognize and recover at the first stall indication.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Slow flight: configuration, coordination and exit · 18 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Recognize and recover at the first stall indication · 24 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Demonstrate one approach-type and one departure-type entry in the approved aircraft configuration. Name the cues as they appear. Recover at the first stall indication by reducing angle of attack and using the POH recovery sequence; do not continue to a full stall in this exercise.

Student exercise: 1. Clear, set the configuration and verify the recovery floor. 2. Enter the instructor-briefed approach scenario. 3. Call the first stall cue and begin recovery immediately. 4. Re-establish stable flight. 5. Repeat for a departure scenario, staying coordinated throughout.

Practice: One instructor example of each entry; one student recovery of each if readiness and altitude allow.

Look for: Names the warning cues and reduces angle of attack without delay; assisted recovery is acceptable on the introduction lesson.

Watch / correct: Trying to climb out by pulling - reduce angle of attack first, then apply approved power.

Watch / correct: Applying aileron aggressively with yaw present - restore coordination and follow the POH stall recovery.

Stop / transition: Instructor intervenes before an uncoordinated stall or spin tendency; recover by the agreed altitude. This lesson does not authorize intentional spins.

Reference: AFH: stall recognition and prevention; POH: stall recovery

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 5, 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 5 & 6

Sporty's course preparation: Vol 2: Segment 7

Sporty's course preparation: Vol 3: Segment 1

Sporty's Academy Private TCO V20: lesson 5 - source page 24 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 6 · Stability, load and wake preparation

Explain stability, coordination, load factor and wake avoidance before more precise control work.

[Return to lesson index](#)

Ground preparation objectives

- Explain stability, coordination, load factor and wake avoidance before more precise control work.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 5](#)

[Current AIM: Chapter 7](#)

[Sporty's course preparation: Vol 2: Segment 15](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 7 · Precise speed, heading and altitude transitions

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	☐ Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	☐ Takeoff, climb and safe transit	1.
16-24 Air 08-16	☐ Airspeed changes and trim	1.
24-36 Air 16-28	☐ Constant-airspeed and constant-rate altitude changes	1.
36-44 Air 28-36	☐ Coordinated turns to headings	1.
44-52 Air 36-44	☐ Slow flight: configuration, coordination and exit	1.
52-58 Air 44-50	☐ Recognize and recover at the first stall indication	1.
58-68 Air 50-60	☐ Return, stabilized approach and final landing	1.
68-72 Ground	☐ Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 7

Link trim and pitch/power to constant-airspeed and selected-rate altitude changes. Revisit slow flight and first-indication stall recovery without progressing to full stalls.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Link trim and pitch/power to constant-airspeed and selected-rate altitude changes. Revisit slow flight and first-indication stall recovery without progressing to full stalls.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Airspeed changes and trim; Constant-airspeed and constant-rate altitude changes; Coordinated turns to headings; Slow flight: configuration, coordination and exit; Recognize and recover at the first stall indication.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Airspeed changes and trim · 8 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Show cruise-to-low-cruise airspeed by setting power, holding altitude with pitch and trimming only after the attitude is established. Reverse the sequence to cruise and demonstrate how trim relieves force without setting the flight path.

Student exercise: 1. Choose the assigned airspeed and configuration within the POH limits. 2. Change power while holding altitude. 3. Stabilize the attitude, then trim. 4. Return to cruise. 5. Explain which control establishes attitude and which relieves pressure.

Practice: Two complete speed transitions, with one hands-light trim check after stabilization.

Look for: Controls altitude through a speed change, avoids chasing trim and verifies the airplane stays stable when control pressure is relaxed.

Watch / correct: Using trim as the primary pitch control - establish attitude with the elevator before trimming.

Watch / correct: Adding power without compensating pitch - anticipate the attitude change and recheck altitude.

Stop / transition: Leave the exercise before stall warning, a configuration limit or an unsafe altitude trend; use the recovery already taught rather than continuing a speed target.

Reference: AFH: straight-and-level flight, transitions and trim

Constant-airspeed and constant-rate altitude changes · 12 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Compare a constant-airspeed climb with a selected-rate climb using aircraft-appropriate power. Show anticipation of level-off. Repeat in descent, emphasizing that rate targets never justify an unsafe airspeed or configuration.

Student exercise: 1. Brief the target altitude and control priorities. 2. Establish the assigned climb mode. 3. Cross-check rate, airspeed and altitude without abandoning lookout. 4. Lead the level-off. 5. Repeat in descent using the other mode and explain the difference.

Practice: One constant-airspeed and one selected-rate climb; one of each descent when spacing and altitude permit.

Look for: Selects the proper pitch/power combination, anticipates level-off and corrects a trend before it becomes a large deviation.

Watch / correct: Fixating on rate while airspeed decays - restore a safe pitch/airspeed and reduce the rate target.

Watch / correct: Leveling late - announce the approaching altitude and begin the transition earlier.

Stop / transition: Level off or restore cruise if assigned airspace, terrain clearance or an airspeed/configuration limit would be approached.

Reference: AFH: climbs and descents; PHAK: instruments

Coordinated turns to headings · 8 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Show clearing, an outside bank reference and coordinated rudder. Compare a shallow turn with a medium-bank turn, then show leading the rollout to capture a heading without skidding.

Student exercise: 1. Clear toward the turn. 2. Enter the briefed bank with coordinated controls. 3. Maintain the outside pitch picture. 4. Anticipate rollout to the chosen heading. 5. Re-trim only when stable and reverse direction after another clearing scan.

Practice: Two turns in each direction, using different heading changes and one deliberate rollout correction.

Look for: Maintains a coordinated turn, identifies altitude drift and rolls out smoothly with decreasing instructor assistance.

Watch / correct: Staring at the heading instrument - use it briefly and return to the outside reference.

Watch / correct: Using rudder alone to hurry the rollout - reduce bank with coordinated aileron and rudder.

Stop / transition: Roll wings level and discontinue if the turn is uncoordinated, traffic appears or the learner loses the altitude/airspeed picture.

Reference: AFH: turns; PHAK: adverse yaw and coordination

Slow flight: configuration, coordination and exit · 8 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Recognize and recover at the first stall indication · 6 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Demonstrate one approach-type and one departure-type entry in the approved aircraft configuration. Name the cues as they appear. Recover at the first stall indication by reducing angle of attack and using the POH recovery sequence; do not continue to a full stall in this exercise.

Student exercise: 1. Clear, set the configuration and verify the recovery floor. 2. Enter the instructor-briefed approach scenario. 3. Call the first stall cue and begin recovery immediately. 4. Re-establish stable flight. 5. Repeat for a departure scenario, staying coordinated throughout.

Practice: One instructor example of each entry; one student recovery of each if readiness and altitude allow.

Look for: Names the warning cues and reduces angle of attack without delay; assisted recovery is acceptable on the introduction lesson.

Watch / correct: Trying to climb out by pulling - reduce angle of attack first, then apply approved power.

Watch / correct: Applying aileron aggressively with yaw present - restore coordination and follow the POH stall recovery.

Stop / transition: Instructor intervenes before an uncoordinated stall or spin tendency; recover by the agreed altitude. This lesson does not authorize intentional spins.

Reference: AFH: stall recognition and prevention; POH: stall recovery

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.

- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- First-indication stall recovery: the six-minute block is a brief review, not a fresh demonstration of both entries. Add another dual block if either entry needs rebuilding.

Study and sources

FAA Airplane Flying Handbook: Chapter 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 5, 6, & 11

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 7 - source page 26 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 8 · Performance preparation

Use the actual aircraft manual for speeds, configuration limits and performance before full-stall/steep-turn practice.

[Return to lesson index](#)

Ground preparation objectives

- Use the actual aircraft manual for speeds, configuration limits and performance before full-stall/steep-turn practice.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Airplane Flying Handbook: Chapters 3 & 6

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 4, 5, 10, & 11

Sporty's course preparation: Vol 3: Segments 10

Sporty's course preparation: Vol 5: Segment 5-7

Sporty's course preparation: Vol 6: Segment 7, 8

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 9 · Full stalls and steep-turn introduction

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-30 Air 08-22	[] Full power-off stalls: approach scenario	1.
30-44 Air 22-36	[] Full power-on stalls: departure scenario	1.
44-58 Air 36-50	[] Steep turns in both directions	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 9

Progress from warning recognition to coordinated full-stall recovery in approved flap configurations, then introduce the steeper-bank control picture.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Progress from warning recognition to coordinated full-stall recovery in approved flap configurations, then introduce the steeper-bank control picture.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Full power-off stalls: approach scenario; Full power-on stalls: departure scenario; Steep turns in both directions.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Full power-off stalls: approach scenario · 14 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: From a safe altitude, show an approach-type full stall with coordinated controls. Identify the first warning and the full-stall indication, then reduce angle of attack, add approved power and clean up in the POH sequence. Contrast the entry with and without flaps.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One entry with flaps and one without on introduction; on review add one briefed turning entry only after straight entries are safe.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 14 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Demonstrate the approved departure configuration and controlled pitch-up at the briefed power. Show the rudder required as speed decreases. At the full stall, reduce angle of attack, maintain coordination and restore a safe climb according to the POH.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One straight entry, then the other approved configuration or a shallow turning entry on a review lesson; never rush the reset.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Steep turns in both directions · 14 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show clearing, the current Private ACS bank target, the added back pressure/power as needed and an outside reference that reveals altitude drift. Demonstrate the rollout and a complete reset before changing direction.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: One demonstration followed by one student turn each way on introduction; two student turns each way on review if safe and unrushed.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapter 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 5

Sporty's course preparation: Vol 3: Segment 3

Sporty's Academy Private TCO V20: lesson 9 - source page 28 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 10 - Weather theory preparation

Connect stability, clouds and hazardous weather to the next flight's conditions and cancellation decisions.

[Return to lesson index](#)

Ground preparation objectives

- Connect stability, clouds and hazardous weather to the next flight's conditions and cancellation decisions.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 4 & 12

Sporty's course preparation: Vol 3: Segments 7-8, 17

Sporty's course preparation: Vol 4: Segment 16

Sporty's course preparation: Vol 5: Segments 4, 12-14

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 11 · Independent maneuver setup and stall review

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-24 Air 08-16	[] Slow flight: configuration, coordination and exit	1.
24-36 Air 16-28	[] Steep turns in both directions	1.
36-47 Air 28-39	[] Full power-off stalls: approach scenario	1.
47-58 Air 39-50	[] Full power-on stalls: departure scenario	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 11

Have the student lead safe setup and recovery. Add briefed turning entries only after straight full-stall recoveries are consistently coordinated.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Have the student lead safe setup and recovery. Add briefed turning entries only after straight full-stall recoveries are consistently coordinated.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Slow flight: configuration, coordination and exit; Steep turns in both directions; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Slow flight: configuration, coordination and exit · 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Steep turns in both directions · 12 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show clearing, the current Private ACS bank target, the added back pressure/power as needed and an outside reference that reveals altitude drift.

Demonstrate the rollout and a complete reset before changing direction.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: One demonstration followed by one student turn each way on introduction; two student turns each way on review if safe and unrushed.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Full power-off stalls: approach scenario · 11 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: From a safe altitude, show an approach-type full stall with coordinated controls. Identify the first warning and the full-stall indication, then reduce angle of attack, add approved power and clean up in the POH sequence. Contrast the entry with and without flaps.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One entry with flaps and one without on introduction; on review add one briefed turning entry only after straight entries are safe.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 11 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Demonstrate the approved departure configuration and controlled pitch-up at the briefed power. Show the rudder required as speed decreases. At the full stall, reduce angle of attack, maintain coordination and restore a safe climb according to the POH.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One straight entry, then the other approved configuration or a shallow turning entry on a review lesson; never rush the reset.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapter 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 5

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 11 - source page 30 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 12 · Weather reports and forecasts

Interpret current weather products and build a flight-specific briefing before ground-reference training.

[Return to lesson index](#)

Ground preparation objectives

- Interpret current weather products and build a flight-specific briefing before ground-reference training.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 13

Current AIM: Chapter 7

Sporty's course preparation: Vol 3: Segments 11-13, 18, 22, 23

Sporty's course preparation: Vol 4: Segments 5, 17, 18, 25

Sporty's course preparation: Vol 5: Segments 15, 20, 22

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 13 · Wind correction and ground-reference introduction

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-21 Air 08-13	[] Read the wind before ground-reference maneuvers	1.
21-33 Air 13-25	[] Rectangular course: wind correction by leg	1.
33-45 Air 25-37	[] S-turns across a road	1.
45-58 Air 37-50	[] Turns around a point	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 13

Teach heading versus ground track, then apply changing bank and drift correction to a rectangle, S-turns and turns around a point.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Teach heading versus ground track, then apply changing bank and drift correction to a rectangle, S-turns and turns around a point.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Read the wind before ground-reference maneuvers; Rectangular course: wind correction by leg; S-turns across a road; Turns around a point.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Read the wind before ground-reference maneuvers · 5 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Use a road or field boundary to compare heading with actual ground track. Show how groundspeed changes around a point and why the bank changes through a maneuver. Select a legal, clear practice area with escape options.

Student exercise: 1. Identify wind direction from available evidence. 2. Compare heading and ground track on two headings. 3. Explain where groundspeed and required bank will be greatest. 4. Select the reference, practice altitude and emergency options before maneuvering.

Practice: Two short wind comparisons, then a verbal prediction before the first ground-reference entry.

Look for: Predicts drift and varying groundspeed and chooses a reference that permits safe, legal maneuvering.

Watch / correct: Assuming heading equals track - compare the airplane with a fixed ground feature.

Watch / correct: Selecting a feature near people or obstacles - choose another area before descending.

Stop / transition: Do not enter if terrain, obstructions, populated areas, airspace or traffic cannot support safe practice at the chosen altitude.

Reference: AFH: ground-reference maneuvers; PHAK: wind correction

Rectangular course: wind correction by leg · 12 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Demonstrate downwind entry, planned lateral spacing and changing bank through each corner. Point out crab on crosswind legs and the difference between heading and track; keep the selected altitude and an outside scan.

Student exercise: 1. Clear and establish the briefed entry. 2. Hold parallel ground tracks with wind correction. 3. Change bank smoothly through each corner to preserve spacing. 4. Call the wind correction on each leg. 5. Exit without descending or tightening toward obstacles.

Practice: One instructor circuit and one student circuit on introduction; two student circuits on review if time allows.

Look for: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn.

Watch / correct: Turning too late on the fast downwind corner - anticipate the corner and use an appropriate coordinated bank.

Watch / correct: Looking continuously at the field - alternate the reference with traffic, horizon and altitude scans.

Stop / transition: Exit toward a safe area for traffic, inadequate obstacle clearance, excessive bank or an unstable altitude trend.

Reference: AFH: rectangular course; Private ACS: Ground Reference Maneuvers

S-turns across a road · 12 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show a suitable reference line, an entry perpendicular to it and changing bank with groundspeed. Demonstrate a wings-level crossing before reversing the turn; explain why equal-looking ground arcs need unequal banks.

Student exercise: 1. Clear and approach the reference as briefed. 2. Begin a coordinated turn that compensates for the wind. 3. Cross wings level and reverse the turn. 4. Adjust bank to create the second comparable arc. 5. Exit and describe the strongest drift correction.

Practice: One paired S-turn demonstration and one student pair on introduction; two student pairs on review.

Look for: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination.

Watch / correct: Holding one bank all the way around - anticipate changing groundspeed and vary bank.

Watch / correct: Rushing the reversal with rudder - level the wings at the reference, then enter the opposite bank together.

Stop / transition: Exit if bank/altitude trend is unsafe or the chosen road has traffic/obstructions that compromise the practice area; maintain legal clearances.

Reference: AFH: S-turns; Private ACS: Ground Reference Maneuvers

Turns around a point · 13 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Select a visible point with clear surrounding airspace. Demonstrate entry and variable bank to hold a constant radius as groundspeed changes. Explain the steepest-bank sector and the need to retain an escape route.

Student exercise: 1. Clear and choose a safe radius and altitude. 2. Enter as briefed. 3. Adjust bank to maintain the ground radius. 4. Keep the ball centered and scan outside. 5. Exit after a full circuit and predict corrections for a second direction.

Practice: One instructor circuit and one student circuit on introduction; one student circuit each way on review.

Look for: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan.

Watch / correct: Tightening toward the point when blown inward - reduce bank and re-establish the planned radius.

Watch / correct: Fixating on the point - scan the horizon and traffic, then check the ground reference briefly.

Stop / transition: Leave the maneuver if the required bank exceeds the practice limit, an escape route is compromised or traffic/terrain clearance becomes uncertain.

Reference: AFH: turns around a point; Private ACS: Ground Reference Maneuvers

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapter 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 5

Sporty's course preparation: Vol 2: Segment 1

Sporty's Academy Private TCO V20: lesson 13 - source page 32 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 14 · Weather decision exercise

Practice a real go/no-go weather briefing and identify explicit continue, divert and cancel gates.

[Return to lesson index](#)

Ground preparation objectives

- Practice a real go/no-go weather briefing and identify explicit continue, divert and cancel gates.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 13](#)

[Current AIM: Chapter 7](#)

[Sporty's course: review the sections related to this flight](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 15 · Ground-reference and maneuver consolidation

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-24 Air 08-16	[] Rectangular course: wind correction by leg	1.
24-32 Air 16-24	[] S-turns across a road	1.
32-40 Air 24-32	[] Turns around a point	1.
40-46 Air 32-38	[] Slow flight: configuration, coordination and exit	1.
46-52 Air 38-44	[] Full power-off stalls: approach scenario	1.
52-58 Air 44-50	[] Full power-on stalls: departure scenario	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 15

Review wind correction and slow/stall/steep-turn skills with student-led setups. This broad source list needs more than one flight if several items still need instruction.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Review wind correction and slow/stall/steep-turn skills with student-led setups. This broad source list needs more than one flight if several items still need instruction.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Rectangular course: wind correction by leg; S-turns across a road; Turns around a point; Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Rectangular course: wind correction by leg · 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Demonstrate downwind entry, planned lateral spacing and changing bank through each corner. Point out crab on crosswind legs and the difference between heading and track; keep the selected altitude and an outside scan.

Student exercise: 1. Clear and establish the briefed entry. 2. Hold parallel ground tracks with wind correction. 3. Change bank smoothly through each corner to preserve spacing. 4. Call the wind correction on each leg. 5. Exit without descending or tightening toward obstacles.

Practice: One instructor circuit and one student circuit on introduction; two student circuits on review if time allows.

Look for: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn.

Watch / correct: Turning too late on the fast downwind corner - anticipate the corner and use an appropriate coordinated bank.

Watch / correct: Looking continuously at the field - alternate the reference with traffic, horizon and altitude scans.

Stop / transition: Exit toward a safe area for traffic, inadequate obstacle clearance, excessive bank or an unstable altitude trend.

Reference: AFH: rectangular course; Private ACS: Ground Reference Maneuvers

S-turns across a road - 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show a suitable reference line, an entry perpendicular to it and changing bank with groundspeed. Demonstrate a wings-level crossing before reversing the turn; explain why equal-looking ground arcs need unequal banks.

Student exercise: 1. Clear and approach the reference as briefed. 2. Begin a coordinated turn that compensates for the wind. 3. Cross wings level and reverse the turn. 4. Adjust bank to create the second comparable arc. 5. Exit and describe the strongest drift correction.

Practice: One paired S-turn demonstration and one student pair on introduction; two student pairs on review.

Look for: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination.

Watch / correct: Holding one bank all the way around - anticipate changing groundspeed and vary bank.

Watch / correct: Rushing the reversal with rudder - level the wings at the reference, then enter the opposite bank together.

Stop / transition: Exit if bank/altitude trend is unsafe or the chosen road has traffic/obstructions that compromise the practice area; maintain legal clearances.

Reference: AFH: S-turns; Private ACS: Ground Reference Maneuvers

Turns around a point - 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Select a visible point with clear surrounding airspace. Demonstrate entry and variable bank to hold a constant radius as groundspeed changes. Explain the steepest-bank sector and the need to retain an escape route.

Student exercise: 1. Clear and choose a safe radius and altitude. 2. Enter as briefed. 3. Adjust bank to maintain the ground radius. 4. Keep the ball centered and scan outside. 5. Exit after a full circuit and predict corrections for a second direction.

Practice: One instructor circuit and one student circuit on introduction; one student circuit each way on review.

Look for: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan.

Watch / correct: Tightening toward the point when blown inward - reduce bank and re-establish the planned radius.

Watch / correct: Fixating on the point - scan the horizon and traffic, then check the ground reference briefly.

Stop / transition: Leave the maneuver if the required bank exceeds the practice limit, an escape route is compromised or traffic/terrain clearance becomes uncertain.

Reference: AFH: turns around a point; Private ACS: Ground Reference Maneuvers

Slow flight: configuration, coordination and exit · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: From a safe altitude, show an approach-type full stall with coordinated controls. Identify the first warning and the full-stall indication, then reduce angle of attack, add approved power and clean up in the POH sequence. Contrast the entry with and without flaps.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One entry with flaps and one without on introduction; on review add one briefed turning entry only after straight entries are safe.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Demonstrate the approved departure configuration and controlled pitch-up at the briefed power. Show the rudder required as speed decreases. At the full stall, reduce angle of attack, maintain coordination and restore a safe climb according to the POH.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One straight entry, then the other approved configuration or a shallow turning entry on a review lesson; never rush the reset.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Steep turns in both directions: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

Study and sources

FAA Airplane Flying Handbook: Chapters 7 & 8

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 14

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 15 - source page 34 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 16 · Emergency preparation

Locate the aircraft emergency procedures and rehearse control-first priorities before simulated failures.

[Return to lesson index](#)

Ground preparation objectives

- Locate the aircraft emergency procedures and rehearse control-first priorities before simulated failures.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 18](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapter 6](#)

[Sporty's course preparation: Vol 3: Segments 5-6](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 17 · Reject, go around and handle an abnormal situation

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-22 Air 08-14	[] Rejected-takeoff decision and controlled simulation	1.
22-32 Air 14-24	[] Recognize an equipment failure and choose a safe action	1.
32-48 Air 24-40	[] Simulated engine failure and emergency approach	1.
48-58 Air 40-50	[] Go-around from a rejected landing	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 17

Build early decisions and an orderly response to equipment problems or simulated engine failure. Brief all simulations before movement and preserve actual aircraft systems.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Build early decisions and an orderly response to equipment problems or simulated engine failure. Brief all simulations before movement and preserve actual aircraft systems.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Rejected-takeoff decision and controlled simulation; Recognize an equipment failure and choose a safe action; Simulated engine failure and emergency approach; Go-around from a rejected landing.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Rejected-takeoff decision and controlled simulation · 6 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Before moving, brief rejection triggers, runway remaining and the aircraft stopping procedure. Demonstrate a reject only in an instructor-selected, prebriefed low-risk condition with adequate runway; otherwise use a stopped/verbal scenario and schedule the flight task later.

Student exercise: 1. State what would require rejection before the roll. 2. Keep directional control during the authorized simulation. 3. Reduce power and stop using the POH procedure. 4. Communicate and clear the runway when safe. 5. Explain why continuing would be unsafe.

Practice: One coached decision rehearsal and, only when suitable, one prebriefed controlled reject. No unannounced high-speed abort.

Look for: Recognizes a rejection trigger promptly and preserves runway alignment while stopping with adequate margin.

Watch / correct: Continuing because the instructor has not spoken - act on the briefed rejection trigger.

Watch / correct: Braking aggressively while drifting - maintain directional control and apply the approved stopping procedure.

Stop / transition: Do not simulate if runway, traffic, contamination or aircraft condition reduces stopping margin. A verbal rehearsal does not count as a completed flight task.

Reference: AFH: takeoffs and emergency procedures; aircraft POH

Recognize an equipment failure and choose a safe action · 10 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Use a verbal or approved simulated indication, such as an electrical or instrument fault. Show how to preserve aircraft control, confirm the indication, use the correct checklist and decide whether to land or divert. Do not disable essential equipment.

Student exercise: 1. Identify the abnormal indication. 2. Keep flying and state the immediate risk. 3. Apply only approved memory actions when applicable, then use the checklist. 4. Reassess systems, weather and landing options. 5. Tell the instructor/ATC what help is needed.

Practice: Two contrasting verbal scenarios, one with a land-soon decision and one requiring checklist diagnosis.

Look for: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes.

Watch / correct: Treating the indication before controlling the airplane - restore control, then diagnose.

Watch / correct: Resetting a breaker or moving a control without authority - consult the aircraft checklist and identify the consequence first.

Stop / transition: End the simulation immediately for a real abnormality, distraction from traffic/terrain or any effect on essential equipment.

Reference: AFH: emergency procedures; POH: systems and abnormal checklists

Simulated engine failure and emergency approach · 16 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Brief how the simulation will start, the minimum recovery condition and who controls power. Demonstrate airspeed, landing-site choice and a plan to the site. Use checklist/communications only after control and the landing plan are established.

Student exercise: 1. Establish the POH glide condition. 2. Select a reachable suitable site and plan the approach. 3. Perform approved checklist actions or verbalize unsafe-to-simulate actions. 4. Brief occupants and communications. 5. Evaluate whether the site remains reachable, then recover at the briefed gate.

Practice: One instructor example on introduction and one student scenario; repeat with a different site only with sufficient altitude and time.

Look for: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan.

Watch / correct: Searching indefinitely for the perfect field - select a suitable reachable option and refine it.

Watch / correct: Trying to stretch the glide - maintain the POH glide condition and revise the landing plan.

Stop / transition: Recover with power at the briefed safe altitude/position or earlier for traffic, terrain, a poor glide plan or aircraft concerns. No surprise engine shutdown or off-airport touchdown.

Reference: AFH: emergency approach; Private ACS: Emergency Approach and Landing; POH

Go-around from a rejected landing · 10 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Brief the unstable-approach gate and demonstrate an early decision to go around. Show approved power, pitch, configuration changes and runway/traffic tracking. Explain why cleaning up before a safe climb can increase sink.

Student exercise: 1. Identify the assigned unstable-approach cue. 2. Announce and initiate the go-around promptly. 3. Apply approved power and establish the climb attitude. 4. Change configuration in the POH sequence. 5. Maintain runway/traffic separation and communicate when workload allows.

Practice: One demonstration on introduction and one student go-around; on review one planned and one student-initiated decision when safe.

Look for: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes.

Watch / correct: Waiting to see whether the landing improves - use the briefed gate and commit to a go-around.

Watch / correct: Retracting flaps while still descending - establish the approved climb and configuration sequence.

Stop / transition: Instructor intervenes for a stall tendency, conflict or runway misalignment; do not manufacture an unsafe approach to create a go-around.

Reference: AFH: approaches and go-arounds; Private ACS: Go-Around/Rejected Landing

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 1, 6, 9, & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 5, & 14

Current AIM: Chapter 7

Sparty's course: review the sections related to this flight

Sparty's Academy Private TCO V20: lesson 17 - source page 36 - Sparty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 18 · Rules, ACS and logbook preparation

Connect applicable regulations, operating guidance and logbook/endorsement rules to the next flights.

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Ground preparation objectives

- Connect applicable regulations, operating guidance and logbook/endorsement rules to the next flights.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Airplane Flying Handbook: Chapters 2 & 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 2

Current Part 61 and applicable Part 91 rules

Current AIM: Introduction & Table of Contents

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course preparation: Vol 3: Segment 26

Sporty's course preparation: Vol 5: Segment 10, 11, 16

Sporty's course preparation: Vol 6: Segment 1, 9, 11

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 19 · Crosswind control, slips and no-flap landing

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-32 Air 08-24	[] Crosswind takeoff, crab and sideslip landing	1.
32-40 Air 24-32	[] Sideslip and forward slip: distinct purposes	1.
40-50 Air 32-42	[] Forward slip to a stabilized landing	1.
50-58 Air 42-50	[] Landing without flaps	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 19

Compare crab and sideslip methods, distinguish sideslip from forward slip and practice an approved no-flap approach. Weather and aircraft restrictions decide which methods are demonstrated.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Compare crab and sideslip methods, distinguish sideslip from forward slip and practice an approved no-flap approach. Weather and aircraft restrictions decide which methods are demonstrated.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Crosswind takeoff, crab and sideslip landing; Sideslip and forward slip: distinct purposes; Forward slip to a stabilized landing; Landing without flaps.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Crosswind takeoff, crab and sideslip landing · 16 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show the wind correction on taxi, takeoff roll and climb. In the pattern compare a crab on approach with a sideslip that aligns the longitudinal axis at touchdown. Explain which method/transition the aircraft permits and the crosswind limits selected for this lesson.

Student exercise: 1. Identify wind and select the briefed technique. 2. Hold runway alignment during takeoff with coordinated controls. 3. Correct drift in the climb. 4. Fly a stabilized approach. 5. Remove drift and align the airplane using the approved method; go around if control margin is inadequate.

Practice: One demonstrated circuit and one student circuit on introduction; add a second student landing only within weather and time limits.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

Sideslip and forward slip: distinct purposes · 8 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: At a safe altitude, demonstrate a sideslip to correct drift while tracking a line, then a forward slip to increase descent with a chosen track. Show approved configuration and indications; compare control inputs and how to return smoothly to coordinated flight.

Student exercise: 1. Verify the aircraft permits the selected slip/configuration. 2. Establish a briefed sideslip and state its drift-control purpose. 3. Return to coordinated flight. 4. Establish a forward slip while holding the planned ground track. 5. Recover before the briefed altitude and explain the difference.

Practice: One instructor example and one student entry of each slip; later approach application requires a separate stabilized-landing exercise.

Look for: Distinguishes drift correction from descent control, stays within aircraft limits and recovers without abrupt yaw or pitch changes.

Watch / correct: Skidding instead of slipping - reduce the control input and have the instructor restore the correct combination.

Watch / correct: Trusting an affected airspeed indication without context - use the POH cautions and the briefed pitch/configuration.

Stop / transition: Discontinue for an aircraft restriction, unapproved flap configuration, unstable speed/track or inadequate recovery height. No slip outside POH guidance.

Reference: AFH: slips; POH: operating limitations and approved procedures

Forward slip to a stabilized landing · 10 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show when a forward slip can correct excess height on an otherwise safe approach. Establish the approved slip/configuration, preserve the planned track and remove the slip early enough to judge a stabilized touchdown.

Student exercise: 1. Identify excess height and confirm the slip is appropriate. 2. Apply the approved forward-slip technique. 3. Hold the landing track and monitor the POH airspeed cautions. 4. Recover to the required landing alignment. 5. Land only if the final approach is stable; otherwise go around.

Practice: One demonstrated approach on introduction and one student approach; do not repeatedly salvage an unstable approach.

Look for: Uses the slip deliberately, exits smoothly and chooses a go-around if a normal safe touchdown is no longer assured.

Watch / correct: Slipping merely to rescue a rushed approach - go around and rebuild the pattern.

Watch / correct: Recovering the slip too late - use the prebriefed stabilization gate and restore alignment earlier.

Stop / transition: Go around for unapproved configuration, excessive sink/drift, uncertainty about airspeed or failure to stabilize before the aircraft/instructor gate.

Reference: AFH: forward slip to landing; Private ACS: Forward Slip to a Landing; POH

Landing without flaps · 8 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Brief the aircraft-approved no-flap approach speed, runway performance and altered sight picture. Demonstrate planning the pattern for the lower drag and avoiding a late, steep correction.

Student exercise: 1. Confirm no-flap performance and runway margin from the POH. 2. Set the approved approach configuration and speed. 3. Judge the higher energy and adjusted descent path. 4. Stabilize and align the aircraft. 5. Land or go around before excess speed/height threatens runway margin.

Practice: One demonstration on introduction and one student no-flap landing when runway and traffic permit.

Look for: Plans the approach early, controls energy and touches down with adequate runway remaining without forcing the airplane onto the runway.

Watch / correct: Using the usual flap-landing picture and arriving fast - adjust pattern and energy earlier.

Watch / correct: Diving to the aiming point - go around and plan a stable descent.

Stop / transition: Go around if approach energy, alignment or remaining runway becomes uncertain; do not simulate a flap failure by creating an actual system fault.

Reference: AFH: no-flap approach; POH: configuration and landing performance

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Rejected-takeoff decision and controlled simulation: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

Study and sources

FAA Airplane Flying Handbook: Chapters 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 2

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 19 - source page 38 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 20 · Aircraft systems preparation I

Use the actual aircraft systems diagrams and POH to explain engine, fuel and electrical operations.

[Return to lesson index](#)

Ground preparation objectives

- Use the actual aircraft systems diagrams and POH to explain engine, fuel and electrical operations.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 3, 6 & 7

Sporty's course preparation: Vol 1: Segments 8-11

Sporty's course preparation: Vol 2: Segments 3, 4

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 21 · Low-speed proficiency and ordinary landing consistency

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-30 Air 08-22	[] Slow flight: configuration, coordination and exit	1.
30-44 Air 22-36	[] Full power-off stalls: approach scenario	1.
44-58 Air 36-50	[] Full power-on stalls: departure scenario	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 21

Ask for student-led slow-flight and full-stall recoveries, then judge whether normal landing control is becoming repeatable without continuous coaching.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Ask for student-led slow-flight and full-stall recoveries, then judge whether normal landing control is becoming repeatable without continuous coaching.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight • 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing • 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Slow flight: configuration, coordination and exit · 14 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 14 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: From a safe altitude, show an approach-type full stall with coordinated controls. Identify the first warning and the full-stall indication, then reduce angle of attack, add approved power and clean up in the POH sequence. Contrast the entry with and without flaps.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One entry with flaps and one without on introduction; on review add one briefed turning entry only after straight entries are safe.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 14 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Demonstrate the approved departure configuration and controlled pitch-up at the briefed power. Show the rudder required as speed decreases. At the full stall, reduce angle of attack, maintain coordination and restore a safe climb according to the POH.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One straight entry, then the other approved configuration or a shallow turning entry on a review lesson; never rush the reset.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 5, 6, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 5 & 14

Current AIM: Chapter 4

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 21 - source page 40 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 22 · Aircraft systems preparation II

Review instruments and avionics limitations and explain what indications would reveal a failure.

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Ground preparation objectives

- Review instruments and avionics limitations and explain what indications would reveal a failure.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 7](#)

[Sporty's course: review the sections related to this flight](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 23 · Presolo landing rehearsal

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-28 Air 08-20	[] Crosswind takeoff, crab and sideslip landing	1.
28-38 Air 20-30	[] Forward slip to a stabilized landing	1.
38-48 Air 30-40	[] Landing without flaps	1.
48-58 Air 40-50	[] Go-around from a rejected landing	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 23

Build repeatable patterns, crosswind decisions, slips, no-flap landing and go-around judgment. Use a normal landing between changed configurations when the student needs a reset.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Build repeatable patterns, crosswind decisions, slips, no-flap landing and go-around judgment. Use a normal landing between changed configurations when the student needs a reset.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Crosswind takeoff, crab and sideslip landing; Forward slip to a stabilized landing; Landing without flaps; Go-around from a rejected landing.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Crosswind takeoff, crab and sideslip landing · 12 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify wind and select the briefed technique. 2. Hold runway alignment during takeoff with coordinated controls. 3. Correct drift in the climb. 4. Fly a stabilized approach. 5. Remove drift and align the airplane using the approved method; go around if control margin is inadequate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

Forward slip to a stabilized landing · 10 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Uses the slip deliberately, exits smoothly and chooses a go-around if a normal safe touchdown is no longer assured. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify excess height and confirm the slip is appropriate. 2. Apply the approved forward-slip technique. 3. Hold the landing track and monitor the POH airspeed cautions. 4. Recover to the required landing alignment. 5. Land only if the final approach is stable; otherwise go around.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Uses the slip deliberately, exits smoothly and chooses a go-around if a normal safe touchdown is no longer assured.

Watch / correct: Slipping merely to rescue a rushed approach - go around and rebuild the pattern.

Watch / correct: Recovering the slip too late - use the prebriefed stabilization gate and restore alignment earlier.

Stop / transition: Go around for unapproved configuration, excessive sink/drift, uncertainty about airspeed or failure to stabilize before the aircraft/instructor gate.

Reference: AFH: forward slip to landing; Private ACS: Forward Slip to a Landing; POH

Landing without flaps · 10 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Plans the approach early, controls energy and touches down with adequate runway remaining without forcing the airplane onto the runway. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Confirm no-flap performance and runway margin from the POH. 2. Set the approved approach configuration and speed. 3. Judge the higher energy and adjusted descent path. 4. Stabilize and align the aircraft. 5. Land or go around before excess speed/height threatens runway margin.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Plans the approach early, controls energy and touches down with adequate runway remaining without forcing the airplane onto the runway.

Watch / correct: Using the usual flap-landing picture and arriving fast - adjust pattern and energy earlier.

Watch / correct: Diving to the aiming point - go around and plan a stable descent.

Stop / transition: Go around if approach energy, alignment or remaining runway becomes uncertain; do not simulate a flap failure by creating an actual system fault.

Reference: AFH: no-flap approach; POH: configuration and landing performance

Go-around from a rejected landing · 10 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify the assigned unstable-approach cue. 2. Announce and initiate the go-around promptly. 3. Apply approved power and establish the climb attitude. 4. Change configuration in the POH sequence. 5. Maintain runway/traffic separation and communicate when workload allows.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes.

Watch / correct: Waiting to see whether the landing improves - use the briefed gate and commit to a go-around.

Watch / correct: Retracting flaps while still descending - establish the approved climb and configuration sequence.

Stop / transition: Instructor intervenes for a stall tendency, conflict or runway misalignment; do not manufacture an unsafe approach to create a go-around.

Reference: AFH: approaches and go-arounds; Private ACS: Go-Around/Rejected Landing

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Solo-readiness requires satisfactory presolo knowledge, required logged training and consistently safe patterns, landings, recoveries and emergency decisions in the intended aircraft. Coaching needed to prevent an unsafe outcome requires further dual instruction.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 14

Current AIM: Chapter 4

Current Private Pilot Airplane ACS: lesson-related tasks

Sparty's course: review the sections related to this flight

Sparty's Academy Private TCO V20: lesson 23 - source page 42 - Sparty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 24 · Systems, instruments and maintenance

Connect instrument indications, maintenance status and pilot responsibilities to presolo readiness.

[Return to lesson index](#)

Ground preparation objectives

- Connect instrument indications, maintenance status and pilot responsibilities to presolo readiness.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 8 & 9

Sporty's course preparation: Vol 3: Segments 9, 14

Sporty's course preparation: Vol 4: Segment 14

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 25 · Presolo knowledge test and flight rehearsal

[Return to lesson index](#)

72 min dual | Brief 20 | Preflight 15 | Debrief 10

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-22 Air 08-14	[] Sight picture, climb, level flight and descent	1.
22-26 Air 14-18	[] Coordinated turns to headings	1.
26-32 Air 18-24	[] Slow flight: configuration, coordination and exit	1.
32-38 Air 24-30	[] Full power-off stalls: approach scenario	1.
38-44 Air 30-36	[] Full power-on stalls: departure scenario	1.
44-49 Air 36-41	[] Recognize an equipment failure and choose a safe action	1.
49-58 Air 41-50	[] Simulated engine failure and emergency approach	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 25

Administer the aircraft/airport/rules presolo test and review every incorrect answer before any solo authorization. Fly a rehearsal emphasizing independent control and a safe emergency landing plan.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Administer the aircraft/airport/rules presolo test and review every incorrect answer before any solo authorization. Fly a rehearsal emphasizing independent control and a safe emergency landing plan.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 20 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Sight picture, climb, level flight and descent; Coordinated turns to headings; Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario; Recognize an equipment failure and choose a safe action; Simulated engine failure and emergency approach.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional

spins.

- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.
- Administer the presolo knowledge test covering applicable rules, local airspace/airport procedures and this aircraft's characteristics/limitations. Grade it and review every wrong answer before solo authorization. The 30-minute source discussion may need extension to finish the test and corrections.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Sight picture, climb, level flight and descent · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Recognizes the changing sight picture, coordinates pitch and power and levels with timely coaching; initial-flight assistance is expected. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Describe the outside attitude and cross-check the instruments. 2. Hold straight flight with coordinated rudder. 3. Establish the briefed climb airspeed and level at the assigned altitude. 4. Establish a descent, then restore level flight. 5. State what pitch and power changed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the changing sight picture, coordinates pitch and power and levels with timely coaching; initial-flight assistance is expected.

Watch / correct: Chasing the vertical-speed indication - return eyes to the horizon and make one small pitch correction.

Watch / correct: Ball displaced during a power change - add the appropriate rudder while keeping wings level.

Stop / transition: Restore ordinary cruise if traffic, turbulence or workload prevents an outside scan; instructor takes control before an unsafe altitude or airspeed develops.

Reference: AFH: basic flight maneuvers; PHAK: flight controls and instruments

Coordinated turns to headings · 4 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains a coordinated turn, identifies altitude drift and rolls out smoothly with decreasing instructor assistance. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Clear toward the turn. 2. Enter the briefed bank with coordinated controls. 3. Maintain the outside pitch picture. 4. Anticipate rollout to the chosen heading. 5. Re-trim only when stable and reverse direction after another clearing scan.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains a coordinated turn, identifies altitude drift and rolls out smoothly with decreasing instructor assistance.

Watch / correct: Staring at the heading instrument - use it briefly and return to the outside reference.

Watch / correct: Using rudder alone to hurry the rollout - reduce bank with coordinated aileron and rudder.

Stop / transition: Roll wings level and discontinue if the turn is uncoordinated, traffic appears or the learner loses the altitude/airspeed picture.

Reference: AFH: turns; PHAK: adverse yaw and coordination

Slow flight: configuration, coordination and exit · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Recognize an equipment failure and choose a safe action · 5 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify the abnormal indication. 2. Keep flying and state the immediate risk. 3. Apply only approved memory actions when applicable, then use the checklist. 4. Reassess systems, weather and landing options. 5. Tell the instructor/ATC what help is needed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes.

Watch / correct: Treating the indication before controlling the airplane - restore control, then diagnose.

Watch / correct: Resetting a breaker or moving a control without authority - consult the aircraft checklist and identify the consequence first.

Stop / transition: End the simulation immediately for a real abnormality, distraction from traffic/terrain or any effect on essential equipment.

Reference: AFH: emergency procedures; POH: systems and abnormal checklists

Simulated engine failure and emergency approach · 9 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Establish the POH glide condition. 2. Select a reachable suitable site and plan the approach. 3. Perform approved checklist actions or verbalize unsafe-to-simulate actions. 4. Brief occupants and communications. 5. Evaluate whether the site remains reachable, then recover at the briefed gate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan.

Watch / correct: Searching indefinitely for the perfect field - select a suitable reachable option and refine it.

Watch / correct: Trying to stretch the glide - maintain the POH glide condition and revise the landing plan.

Stop / transition: Recover with power at the prebriefed safe altitude/position or earlier for traffic, terrain, a poor glide plan or aircraft concerns. No surprise engine shutdown or off-airport touchdown.

Reference: AFH: emergency approach; Private ACS: Emergency Approach and Landing; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Solo-readiness requires satisfactory presolo knowledge, required logged training and consistently safe patterns, landings, recoveries and emergency decisions in the intended aircraft. Coaching needed to prevent an unsafe outcome requires further dual instruction.

Debrief · 10 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Steep turns in both directions: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Sideslip and forward slip: distinct purposes: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Landing without flaps: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 5, 6, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 5, & 14

Current AIM: Chapter 4

Sporty's course preparation: Vol 1: Review Segments as Needed

Sporty's course preparation: Vol 2: Review Segments as Needed

Sporty's course preparation: Vol 3: Segments 26-28; Review Segments as Needed

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Lesson 26 - Airspace preparation

Identify the home/practice-area airspace, entry requirements and weather minima before solo-readiness assessment.

[Return to lesson index](#)

Ground preparation objectives

- Identify the home/practice-area airspace, entry requirements and weather minima before solo-readiness assessment.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 15](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapter 3](#)

[Sporty's course preparation: Vol 4: Segments 22, 23](#)

[Sporty's course preparation: Vol 6: Segment 3](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 27 · Stage I: independent solo-readiness assessment

[Return to lesson index](#)

72 min check | Brief 60 | Preflight 15 | Debrief 30

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-26 Air 08-18	[] Slow flight: configuration, coordination and exit	1.
26-34 Air 18-26	[] Full power-off stalls: approach scenario	1.
34-42 Air 26-34	[] Full power-on stalls: departure scenario	1.
42-48 Air 34-40	[] Recognize an equipment failure and choose a safe action	1.
48-58 Air 40-50	[] Simulated engine failure and emergency approach	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 27

Assess the knowledge and flight skills needed for safe solo operation. Record any assistance or unobserved task and send it back to training before a solo release.

Source clarification: The lesson page specifies 72 minutes of flight; the publisher allocation table lists 90. This plan uses the lesson-page target and does not claim those totals agree.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Assess the knowledge and flight skills needed for safe solo operation. Record any assistance or unobserved task and send it back to training before a solo release.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Finish required training and prepare an assessor plan of action. Identify prior deficiencies and unobserved tasks before the assessment.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 60 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario; Recognize an equipment failure and choose a safe action; Simulated engine failure and emergency approach.

- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.
- Use the parked oral to assess certificates/endorsements, airworthiness, current weather, aircraft performance/systems, airspace and human factors. Include airport signs/markings/lighting and emergency/survival equipment where in scope. Final assessments also cover the planned cross-country, night preparation and knowledge-test deficiency codes.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Slow flight: configuration, coordination and exit · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Recognize an equipment failure and choose a safe action · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Identify the abnormal indication. 2. Keep flying and state the immediate risk. 3. Apply only approved memory actions when applicable, then use the checklist. 4. Reassess systems, weather and landing options. 5. Tell the instructor/ATC what help is needed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes.

Watch / correct: Treating the indication before controlling the airplane - restore control, then diagnose.

Watch / correct: Resetting a breaker or moving a control without authority - consult the aircraft checklist and identify the consequence first.

Stop / transition: End the simulation immediately for a real abnormality, distraction from traffic/terrain or any effect on essential equipment.

Reference: AFH: emergency procedures; POH: systems and abnormal checklists

Simulated engine failure and emergency approach · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Establish the POH glide condition. 2. Select a reachable suitable site and plan the approach. 3. Perform approved checklist actions or verbalize unsafe-to-simulate actions. 4. Brief occupants and communications. 5. Evaluate whether the site remains reachable, then recover at the briefed gate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan.

Watch / correct: Searching indefinitely for the perfect field - select a suitable reachable option and refine it.

Watch / correct: Trying to stretch the glide - maintain the POH glide condition and revise the landing plan.

Stop / transition: Recover with power at the prebriefed safe altitude/position or earlier for traffic, terrain, a poor glide plan or aircraft concerns. No surprise engine shutdown or off-airport touchdown.

Reference: AFH: emergency approach; Private ACS: Emergency Approach and Landing; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Solo-readiness requires satisfactory presolo knowledge, required logged training and consistently safe patterns, landings, recoveries and emergency decisions in the intended aircraft. Coaching needed to prevent an unsafe outcome requires further dual instruction.

Debrief · 30 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

Current Private Pilot Airplane ACS: knowledge, risk and skill review

Sporty's Academy Private TCO V20: lesson 27 - source page 47 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 28 · Charts and publications preparation

Find current sectional/airport information and recognize features needed for later pilotage flights.

[Return to lesson index](#)

Ground preparation objectives

- Find current sectional/airport information and recognize features needed for later pilotage flights.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[Sporty's course preparation: Vol 4: Segments 3, 4, 6](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 29 · Solo-readiness patterns and go-around decisions

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-40 Air 08-32	[] Crosswind takeoff, crab and sideslip landing	1.
40-58 Air 32-50	[] Go-around from a rejected landing	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 29

Reduce coaching and let the student make pattern, traffic and go-around decisions. Evaluate each landing as a complete arrival-to-rollout sequence.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Reduce coaching and let the student make pattern, traffic and go-around decisions. Evaluate each landing as a complete arrival-to-rollout sequence.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Crosswind takeoff, crab and sideslip landing; Go-around from a rejected landing.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Crosswind takeoff, crab and sideslip landing · 24 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify wind and select the briefed technique. 2. Hold runway alignment during takeoff with coordinated controls. 3. Correct drift in the climb. 4. Fly a stabilized approach. 5. Remove drift and align the airplane using the approved method; go around if control margin is inadequate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

Go-around from a rejected landing · 18 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify the assigned unstable-approach cue. 2. Announce and initiate the go-around promptly. 3. Apply approved power and establish the climb attitude. 4. Change configuration in the POH sequence. 5. Maintain runway/traffic separation and communicate when workload allows.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes.

Watch / correct: Waiting to see whether the landing improves - use the briefed gate and commit to a go-around.

Watch / correct: Retracting flaps while still descending - establish the approved climb and configuration sequence.

Stop / transition: Instructor intervenes for a stall tendency, conflict or runway misalignment; do not manufacture an unsafe approach to create a go-around.

Reference: AFH: approaches and go-arounds; Private ACS: Go-Around/Rejected Landing

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Solo-readiness requires satisfactory presolo knowledge, required logged training and consistently safe patterns, landings, recoveries and emergency decisions in the intended aircraft. Coaching needed to prevent an unsafe outcome requires further dual instruction.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 5, 6, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 5, & 14

Current AIM: Chapter 4

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 29 - source page 49 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 30 - Human factors preparation

Prepare personal readiness, impairment, disorientation and workload decisions before the first solo gate.

[Return to lesson index](#)

Ground preparation objectives

- Prepare personal readiness, impairment, disorientation and workload decisions before the first solo gate.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 17](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapter 8](#)

[Sporty's course preparation: Vol 3: Segment 27](#)

[Sporty's course preparation: Vol 6: Segment 4](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 31 · Final dual gate before first solo

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-36 Air 08-28	[] Crosswind takeoff, crab and sideslip landing	1.
36-42 Air 28-34	[] Recognize an equipment failure and choose a safe action	1.
42-52 Air 34-44	[] Simulated engine failure and emergency approach	1.
52-58 Air 44-50	[] Go-around from a rejected landing	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 31

Require safe patterns without coaching, plus an independent response to a briefed abnormality and simulated emergency approach. Do not schedule first solo because a lesson number has been reached.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Require safe patterns without coaching, plus an independent response to a briefed abnormality and simulated emergency approach. Do not schedule first solo because a lesson number has been reached.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Crosswind takeoff, crab and sideslip landing; Recognize an equipment failure and choose a safe action; Simulated engine failure and emergency approach; Go-around from a rejected landing.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Crosswind takeoff, crab and sideslip landing · 20 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify wind and select the briefed technique. 2. Hold runway alignment during takeoff with coordinated controls. 3. Correct drift in the climb. 4. Fly a stabilized approach. 5. Remove drift and align the airplane using the approved method; go around if control margin is inadequate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

Recognize an equipment failure and choose a safe action · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify the abnormal indication. 2. Keep flying and state the immediate risk. 3. Apply only approved memory actions when applicable, then use the checklist. 4. Reassess systems, weather and landing options. 5. Tell the instructor/ATC what help is needed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Prioritizes aircraft control, chooses the correct checklist and reaches a defensible landing/diversion decision without random switch changes.

Watch / correct: Treating the indication before controlling the airplane - restore control, then diagnose.

Watch / correct: Resetting a breaker or moving a control without authority - consult the aircraft checklist and identify the consequence first.

Stop / transition: End the simulation immediately for a real abnormality, distraction from traffic/terrain or any effect on essential equipment.

Reference: AFH: emergency procedures; POH: systems and abnormal checklists

Simulated engine failure and emergency approach · 10 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Establish the POH glide condition. 2. Select a reachable suitable site and plan the approach. 3. Perform approved checklist actions or verbalize unsafe-to-simulate actions. 4. Brief occupants and communications. 5. Evaluate whether the site remains reachable, then recover at the briefed gate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains glide control, selects a reachable site early and manages checklist/communications without losing the landing plan.

Watch / correct: Searching indefinitely for the perfect field - select a suitable reachable option and refine it.

Watch / correct: Trying to stretch the glide - maintain the POH glide condition and revise the landing plan.

Stop / transition: Recover with power at the prebriefed safe altitude/position or earlier for traffic, terrain, a poor glide plan or aircraft concerns. No surprise engine shutdown or off-airport touchdown.

Reference: AFH: emergency approach; Private ACS: Emergency Approach and Landing; POH

Go-around from a rejected landing · 6 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify the assigned unstable-approach cue. 2. Announce and initiate the go-around promptly. 3. Apply approved power and establish the climb attitude. 4. Change configuration in the POH sequence. 5. Maintain runway/traffic separation and communicate when workload allows.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Makes the decision early, controls yaw/pitch and transitions to a safe climb without premature configuration changes.

Watch / correct: Waiting to see whether the landing improves - use the briefed gate and commit to a go-around.

Watch / correct: Retracting flaps while still descending - establish the approved climb and configuration sequence.

Stop / transition: Instructor intervenes for a stall tendency, conflict or runway misalignment; do not manufacture an unsafe approach to create a go-around.

Reference: AFH: approaches and go-arounds; Private ACS: Go-Around/Rejected Landing

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Solo-readiness requires satisfactory presolo knowledge, required logged training and consistently safe patterns, landings, recoveries and emergency decisions in the intended aircraft. Coaching needed to prevent an unsafe outcome requires further dual instruction.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 6, 8, 9, & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2 & 14

Current AIM: Chapters 4 & 6

Current Private Pilot Airplane ACS: lesson-related tasks

Sparty's course: review the sections related to this flight

Sparty's Academy Private TCO V20: lesson 31 - source page 51 - Sparty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 32 · First solo: dual gate, parked release and supervised patterns

[Return to lesson index](#)

96 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] DUAL · Start, taxi, run-up and runway confirmation	1.
08-14 Air 00-06	[] DUAL · Takeoff, climb and safe transit	1.
14-46 Air 06-38	[] DUAL · Crosswind takeoff, crab and sideslip landing	1.
46-56 Air 38-48	[] DUAL · Return, stabilized approach and final landing	1.
56-60 Ground	[] DUAL · Clear, taxi in, shut down and secure	1.
60-66 Ground	[] SOLO · Start, taxi, run-up and runway confirmation	1.
66-71 Air 48-53	[] SOLO · Takeoff, climb and safe transit	1.
71-85 Air 53-67	[] SOLO · Supervised traffic-pattern circuits	1.
85-92 Air 67-74	[] SOLO · Return, stabilized approach and final landing	1.
92-96 Ground	[] SOLO · Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 32

A compound mission: 60 minutes dual first, then a separate 36-minute supervised solo only after a parked instructor release. The source dual and solo targets remain separate.

Source clarification: Source flight target: 60 minutes dual plus 36 minutes solo. The parked instructor release and endorsement/weather recheck between portions are outside flight time and may extend the 12-minute source discussion. Stop after dual if any solo gate is unmet.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- A compound mission: 60 minutes dual first, then a separate 36-minute supervised solo only after a parked instructor release. The source dual and solo targets remain separate.
- Conduct only the specifically authorized route/maneuvers and return within the agreed limits; report actual repetitions and any difficulty.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMS, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: .
- Before release, verify the student certificate/medical eligibility as applicable, current make/model solo endorsement, specific area/airport/route authority, weather/crosswind limits, fuel, aircraft status and

contact/late-return plan. Solo is conditional on demonstrated proficiency; a calendar or timer never releases the student.

- Rehearse real lost/weather/equipment responses before departure. Do not create a lost condition, abnormal system state or unfamiliar maneuver alone; seek help or return when the authorization/safety margin changes.
- After the 60-minute dual portion, shut down and conduct a parked release check. Verify independent patterns/landings, current conditions and endorsements; brief the lighter solo handling, authorized circuits, go-around/early-return triggers and ground supervision. Restart for the separate 36-minute solo only after authorization. If the gate is not met, remain dual and reschedule; the solo timer is not permission.

DUAL · Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

DUAL · Takeoff, climb and safe transit · 6 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

DUAL · Crosswind takeoff, crab and sideslip landing · 32 minutes

Teach / coach: Before flight, assign the setup and recovery/termination gate. Observe without coaching and judge this task-specific result: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance. Intervene for safety, document the assistance and return the task to dual training; a later demonstration is a separate training attempt.

Student exercise: 1. Identify wind and select the briefed technique. 2. Hold runway alignment during takeoff with coordinated controls. 3. Correct drift in the climb. 4. Fly a stabilized approach. 5. Remove drift and align the airplane using the approved method; go around if control margin is inadequate.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

DUAL · Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

DUAL · Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

SOLO · Start, taxi, run-up and runway confirmation · 6 minutes

Teach / coach: Ground supervision: before release verify current solo authority, route/area, weather, fuel, aircraft condition and contact/return plan. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

SOLO · Takeoff, climb and safe transit · 5 minutes

Teach / coach: Ground supervision: observe the authorized departure if practical and retain the agreed contact plan. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

SOLO · Supervised traffic-pattern circuits · 14 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Fly only the released pattern/runway in the briefed weather. 2. Make required calls and maintain separation. 3. Use the normal approved takeoff and landing technique, with wind correction only within the endorsement limits. 4. Go around whenever unstable. 5. End the circuits early if traffic, weather, workload or fuel changes.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Maintains centerline, manages drift and lands aligned without side loading; new crosswind technique may require assistance.

Watch / correct: Holding a fixed correction as airspeed changes - adjust control continuously for the changing effectiveness.

Watch / correct: Touching down while drifting or crabbed beyond aircraft guidance - go around and reset the approach.

Stop / transition: Go around for excessive drift, misalignment, gusts or inadequate control margin. Use aircraft, instructor and endorsement limits; a published demonstrated crosswind is not automatically a legal operating limit.

Reference: AFH: crosswind takeoffs and landings; POH

SOLO · Return, stabilized approach and final landing · 7 minutes

Teach / coach: Ground supervision: remain reachable, monitor the agreed return window and accept an early return/go-around without pressure to finish repetitions. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

SOLO · Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Ground supervision: meet/debrief after shutdown and verify the actual route, landing counts and any difficulty before another authorization. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- The instructor makes and documents the parked release independently of the dual time target; the student then completes a safe supervised solo pattern flight within the actual authorization. If solo does not occur, Stage I is not completed by the dual portion alone.

Debrief · 4 minutes

- Reconstruct the actual route/circuits, landing counts, weather/traffic and any deviation from the authorization.
- Identify any point where help, a go-around, return or diversion was needed; rehearse the response before another solo release.

- Confirm the airplane is secured and reconcile actual experience and next authorization with the instructor outside this app.

Study and sources

FAA Airplane Flying Handbook: Chapters 2, 6, 8, 9, & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2 & 14

Current AIM: Chapters 4 & 6

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's Academy Private TCO V20: lesson 32 - source page 52 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Field performance and visual navigation

Short/soft-field procedures, authorized solo practice, pilotage trips, alternatives and a navigation/performance assessment.

Lesson 33 · Principles of navigation

Prepare heading/course references, visual position fixes and wind correction before later pilotage flights.

[Return to lesson index](#)

Ground preparation objectives

- Prepare heading/course references, visual position fixes and wind correction before later pilotage flights.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 7](#)

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16](#)

[Sporty's course preparation: Vol 5: Segment 3](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 34 · Short- and soft-field technique introduction

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-37 Air 08-29	[] Short-field takeoff and landing	1.
37-58 Air 29-50	[] Soft-field takeoff and landing	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 34

Connect runway condition and POH performance to short-field obstacle clearance and soft-field ground-effect/gear protection. Use a suitable runway to simulate the condition.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Connect runway condition and POH performance to short-field obstacle clearance and soft-field ground-effect/gear protection. Use a suitable runway to simulate the condition.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Short-field takeoff and landing; Soft-field takeoff and landing.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Short-field takeoff and landing · 21 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Use POH performance, actual runway conditions and a selected obstacle/aiming point. Demonstrate the approved takeoff setup, obstacle-clearance climb and transition. On return show stabilized approach speed, precise touchdown and approved stopping technique.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review if the runway/pattern supports them.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 21 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Explain how to unload the nosewheel and preserve momentum using the POH technique. Demonstrate the ground-effect transition without an early climb at insufficient airspeed. On landing show energy management and protection of the landing gear.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review when safe.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 6 & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 11

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course preparation: Vol 5: Segment 9

Sporty's Academy Private TCO V20: lesson 34 - source page 55 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 35 · Publications and flight-planning equipment

Prepare current charts, Chart Supplement, navigation logs and planning tools before cross-country work.

[Return to lesson index](#)

Ground preparation objectives

- Prepare current charts, Chart Supplement, navigation logs and planning tools before cross-country work.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 7, 9, 14, & 16

Current Part 61 and applicable Part 91 rules

Current AIM: Chapter 9

Sporty's course preparation: Vol 4: Segments 7, 8, 10, 19, 24

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 36 · Maximum-performance and low-speed review

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-28 Air 08-20	[] Short-field takeoff and landing	1.
28-40 Air 20-32	[] Soft-field takeoff and landing	1.
40-46 Air 32-38	[] Slow flight: configuration, coordination and exit	1.
46-52 Air 38-44	[] Full power-off stalls: approach scenario	1.
52-58 Air 44-50	[] Full power-on stalls: departure scenario	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 36

Develop smoother short/soft-field performance, then check slow flight and full-stall recovery. Carry the landing slip onto another session if adequate repetitions cannot fit.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Develop smoother short/soft-field performance, then check slow flight and full-stall recovery. Carry the landing slip onto another session if adequate repetitions cannot fit.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Short-field takeoff and landing; Soft-field takeoff and landing; Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Short-field takeoff and landing · 12 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Use POH performance, actual runway conditions and a selected obstacle/aiming point. Demonstrate the approved takeoff setup, obstacle-clearance climb and transition. On return show stabilized approach speed, precise touchdown and approved stopping technique.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review if the runway/pattern supports them.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 12 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Explain how to unload the nosewheel and preserve momentum using the POH technique. Demonstrate the ground-effect transition without an early climb at insufficient airspeed. On landing show energy management and protection of the landing gear.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review when safe.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Slow flight: configuration, coordination and exit · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: From a safe altitude, show an approach-type full stall with coordinated controls. Identify the first warning and the full-stall indication, then reduce angle of attack, add approved power and clean up in the POH sequence. Contrast the entry with and without flaps.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One entry with flaps and one without on introduction; on review add one briefed turning entry only after straight entries are safe.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Demonstrate the approved departure configuration and controlled pitch-up at the briefed power. Show the rudder required as speed decreases. At the full stall, reduce angle of attack, maintain coordination and restore a safe climb according to the POH.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One straight entry, then the other approved configuration or a shallow turning entry on a review lesson; never rush the reset.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

Study and sources

FAA Airplane Flying Handbook: Chapters 5, 6, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 11

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 36 - source page 57 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 37 · Cross-country planning preparation

Prepare route, altitude, weather, heading/time/fuel estimates and landing alternatives before pilotage training.

[Return to lesson index](#)

Ground preparation objectives

- Prepare route, altitude, weather, heading/time/fuel estimates and landing alternatives before pilotage training.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 9 & 16

Current Part 61 and applicable Part 91 rules

Current AIM: Chapters 1 & 9

Sporty's course preparation: Vol 4: Segment 20

Sporty's course preparation: Vol 5: Segment 5

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 38 · Authorized solo practice: maneuver and landing priorities

[Return to lesson index](#)

60 min solo | Brief 12 | Preflight 15 | Debrief 8

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-24 Air 08-16	[] Slow flight: configuration, coordination and exit	1.
24-32 Air 16-24	[] Steep turns in both directions	1.
32-39 Air 24-31	[] Turns around a point	1.
39-46 Air 31-38	[] Short-field takeoff and landing	1.
46-56 Air 38-48	[] Return, stabilized approach and final landing	1.
56-60 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 38

Choose only already-proficient maneuvers specifically authorized for this aircraft, area and weather. The source contains more items than can all receive useful practice in one hour.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Choose only already-proficient maneuvers specifically authorized for this aircraft, area and weather. The source contains more items than can all receive useful practice in one hour.
- Conduct only the specifically authorized route/maneuvers and return within the agreed limits; report actual repetitions and any difficulty.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 12 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Slow flight: configuration, coordination and exit; Steep turns in both directions; Turns around a point; Short-field takeoff and landing.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.

- Before release, verify the student certificate/medical eligibility as applicable, current make/model solo endorsement, specific area/airport/route authority, weather/crosswind limits, fuel, aircraft status and contact/late-return plan. Solo is conditional on demonstrated proficiency; a calendar or timer never releases the student.
- Rehearse real lost/weather/equipment responses before departure. Do not create a lost condition, abnormal system state or unfamiliar maneuver alone; seek help or return when the authorization/safety margin changes.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Ground supervision: before release verify current solo authority, route/area, weather, fuel, aircraft condition and contact/return plan. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Ground supervision: observe the authorized departure if practical and retain the agreed contact plan. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Slow flight: configuration, coordination and exit • 8 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Steep turns in both directions • 8 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Turns around a point • 7 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Clear and choose a safe radius and altitude. 2. Enter as briefed. 3. Adjust bank to maintain the ground radius. 4. Keep the ball centered and scan outside. 5. Exit after a full circuit and predict corrections for a second direction.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan.

Watch / correct: Tightening toward the point when blown inward - reduce bank and re-establish the planned radius.

Watch / correct: Fixating on the point - scan the horizon and traffic, then check the ground reference briefly.

Stop / transition: Leave the maneuver if the required bank exceeds the practice limit, an escape route is compromised or traffic/terrain clearance becomes uncertain.

Reference: AFH: turns around a point; Private ACS: Ground Reference Maneuvers

Short-field takeoff and landing · 7 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Ground supervision: remain reachable, monitor the agreed return window and accept an early return/go-around without pressure to finish repetitions. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Ground supervision: meet/debrief after shutdown and verify the actual route, landing counts and any difficulty before another authorization. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student safely completes the specifically authorized mission and reports actual route, repetitions, full stops and any unplanned assistance/early return. The instructor verifies the required source mission rather than assuming it from flight time.

Debrief · 8 minutes

- Reconstruct the actual route/circuits, landing counts, weather/traffic and any deviation from the authorization.
- Identify any point where help, a go-around, return or diversion was needed; rehearse the response before another solo release.

- Confirm the airplane is secured and reconcile actual experience and next authorization with the instructor outside this app.

Repeat or carry forward

- Full power-off stalls: approach scenario: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Full power-on stalls: departure scenario: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Rectangular course: wind correction by leg: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- S-turns across a road: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Soft-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

Study and sources

FAA Airplane Flying Handbook: Chapters 5-7 & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 5

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 38 - source page 59 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 39 - Navigation preparation II

Build the actual route, airport data, heading/time/fuel estimates and weather decisions for the first pilotage trip.

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Ground preparation objectives

- Build the actual route, airport data, heading/time/fuel estimates and weather decisions for the first pilotage trip.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapter 1](#)

[Sporty's course preparation: Vol 4: Segment 9](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 40 · Pilotage route with two unfamiliar-airport landings

[Return to lesson index](#)

90 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-42 Air 08-34	[] Navigate by landmarks and magnetic compass	1.
42-60 Air 34-52	[] Arrive, land and depart at an unfamiliar airport	1.
60-76 Air 52-68	[] Navigate by landmarks and magnetic compass	1.
76-86 Air 68-78	[] Return, stabilized approach and final landing	1.
86-90 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 40

Plan a visual-navigation trip to an airport more than 25 but less than 50 NM away, plus a landing at another airport within 25 NM of the home airport. Navigate by landmarks and compass.

Source clarification: The 90-minute target is a planning estimate. Verify the actual route, two required airport visits, wind, traffic and fuel; extend the flight or divide training when the safe route needs more time.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Plan a visual-navigation trip to an airport more than 25 but less than 50 NM away, plus a landing at another airport within 25 NM of the home airport. Navigate by landmarks and compass.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Arrive, land and depart at an unfamiliar airport; Navigate by landmarks and magnetic compass.
- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 26 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show the next prominent checkpoint, chart-to-ground orientation and a heading checked with the compass in stable flight. Compare planned and actual time/fuel. Demonstrate a concise position report or Flight Service contact when useful, without letting communications replace lookout.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: At least three checkpoint fixes and two time/fuel updates; use the source route distances rather than a generic local circuit.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Arrive, land and depart at an unfamiliar airport • 18 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Before arrival, review current runway, pattern, traffic and taxi information. Show how to identify the airport/runway visually, make proper non-towered reports or follow a clearance, then stop at a safe place to review departure information.

Student exercise: 1. Confirm destination position, runway/wind, published pattern and landing performance. 2. Establish radio awareness and visually identify traffic. 3. Join the correct pattern and fly a stable approach. 4. Land, clear and complete checks before heads-down taxi work. 5. Review departure/taxi routing before taking off again.

Practice: One complete arrival-landing-departure at each required destination; do not omit a source-required airport to meet the clock.

Look for: Finds the correct airport/runway, integrates safely with traffic and handles taxi/departure without instructor navigation rescue.

Watch / correct: Descending toward a guessed airport - remain safe and verify the airport before committing.

Watch / correct: Programming or folding charts while crossing a runway - stop at a safe location and perform the task there.

Stop / transition: Go around or divert if airport identity, runway suitability, traffic or weather is uncertain; obtain any required clearance before entering controlled movement/runway areas.

Reference: AIM: airport operations; current Chart Supplement, airport diagram and POH

Navigate by landmarks and magnetic compass • 16 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show the next prominent checkpoint, chart-to-ground orientation and a heading checked with the compass in stable flight. Compare planned and actual time/fuel. Demonstrate a concise position report or Flight Service contact when useful, without letting communications replace lookout.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: At least three checkpoint fixes and two time/fuel updates; use the source route distances rather than a generic local circuit.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 14, & 16

Current AIM: Chapters 1, 2, 4, & 9

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 40 - source page 61 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 41 · Flight planning and weather decisions

Prepare alternative airports, changing-weather decisions and en-route estimates for the diversion/lost lesson.

[Return to lesson index](#)

Ground preparation objectives

- Prepare alternative airports, changing-weather decisions and en-route estimates for the diversion/lost lesson.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16

Current Part 61 and applicable Part 91 rules

Current AIM: Chapters 1, 6, & 9

Sporty's course preparation: Vol 4: Segment 1

Sporty's course preparation: Vol 5: Segment 21

Sporty's course preparation: Vol 6: Segment 2

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 42 · Pilotage, diversion and uncertain-position recovery

[Return to lesson index](#)

108 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-41 Air 08-33	[] Navigate by landmarks and magnetic compass	1.
41-56 Air 33-48	[] Arrive, land and depart at an unfamiliar airport	1.
56-74 Air 48-66	[] Choose and fly a diversion	1.
74-84 Air 66-76	[] Recover an uncertain position	1.
84-94 Air 76-86	[] Emergency descent: controlled escape from an urgent hazard	1.
94-104 Air 86-96	[] Return, stabilized approach and final landing	1.
104-108 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 42

Fly toward an airport more than 25 NM away, then practice an alternate-airport decision and a controlled position-recovery scenario. Introduce emergency descent only with ample clear recovery airspace.

Source clarification: The 108-minute publisher target does not establish an actual route duration. Plan and extend the trip for weather, distance and airport operations rather than abbreviating a required arrival or safe recovery.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Fly toward an airport more than 25 NM away, then practice an alternate-airport decision and a controlled position-recovery scenario. Introduce emergency descent only with ample clear recovery airspace.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Arrive, land and depart at an unfamiliar airport; Choose and fly a diversion; Recover an uncertain position; Emergency descent: controlled escape from an urgent hazard.
- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 25 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Show the next prominent checkpoint, chart-to-ground orientation and a heading checked with the compass in stable flight. Compare planned and actual time/fuel. Demonstrate a concise position report or Flight Service contact when useful, without letting communications replace lookout.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: At least three checkpoint fixes and two time/fuel updates; use the source route distances rather than a generic local circuit.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Arrive, land and depart at an unfamiliar airport · 15 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Before arrival, review current runway, pattern, traffic and taxi information. Show how to identify the airport/runway visually, make proper non-towered reports or follow a clearance, then stop at a safe place to review departure information.

Student exercise: 1. Confirm destination position, runway/wind, published pattern and landing performance. 2. Establish radio awareness and visually identify traffic. 3. Join the correct pattern and fly a stable approach. 4. Land, clear and complete checks before heads-down taxi work. 5. Review departure/taxi routing before taking off again.

Practice: One complete arrival-landing-departure at each required destination; do not omit a source-required airport to meet the clock.

Look for: Finds the correct airport/runway, integrates safely with traffic and handles taxi/departure without instructor navigation rescue.

Watch / correct: Descending toward a guessed airport - remain safe and verify the airport before committing.

Watch / correct: Programming or folding charts while crossing a runway - stop at a safe location and perform the task there.

Stop / transition: Go around or divert if airport identity, runway suitability, traffic or weather is uncertain; obtain any required clearance before entering controlled movement/runway areas.

Reference: AIM: airport operations; current Chart Supplement, airport diagram and POH

Choose and fly a diversion · 18 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Present a briefed weather, fuel or destination problem without creating a real hazard. Show selecting a suitable alternative, estimating heading/time/fuel, checking terrain/airspace and changing the flight plan or informing ATC as appropriate.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One complete diversion decision and two position/ETA updates along the new course.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Recover an uncertain position · 10 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Use an agreed simulated uncertainty while retaining a known safe position. Show stabilizing the airplane, using time/heading/landmarks, identifying a safe navigation aid and requesting assistance early. Discuss climb only when weather, terrain and airspace permit.

Student exercise: 1. Admit uncertainty promptly. 2. Maintain control and a safe altitude/course. 3. Check last known position, elapsed time and fuel. 4. Use appropriate navigation/ATC assistance to establish position. 5. Choose a safe landing plan rather than wandering.

Practice: One deliberate position-recovery exercise with a clear ending fix; solo lessons rehearse this before departure and use it only for real uncertainty.

Look for: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace.

Watch / correct: Continuing random turns while searching - stabilize the airplane and work from the last known fix.

Watch / correct: Delaying a radio request through embarrassment - state uncertainty and request assistance early.

Stop / transition: Stop any simulation before actual uncertainty occurs; for a real lost condition, prioritize safety and obtain assistance. Never intentionally get lost on solo.

Reference: PHAK: lost procedures; AIM: assistance and communications

Emergency descent: controlled escape from an urgent hazard · 10 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Brief a simulated smoke/fire or other urgent descent need using the POH procedure. Verify clear airspace, terrain and recovery altitude. Demonstrate approved configuration, bank, airspeed and checklist priorities without exceeding any limitation.

Student exercise: 1. State the simulated hazard and approved immediate actions. 2. Clear the descent path. 3. Establish the POH emergency-descent procedure. 4. Monitor airspeed, load and terrain. 5. Recover by the briefed safe altitude and decide where to land.

Practice: One demonstration on introduction and one student descent if recovery margin is ample; one independently flown scenario on assessment.

Look for: Uses the correct aircraft procedure, descends promptly within limits and retains terrain/traffic awareness through recovery.

Watch / correct: Treating maximum descent rate as the only goal - prioritize aircraft limits and terrain clearance.

Watch / correct: Missing the recovery altitude while checking a list - keep the descent path and altitude in the scan.

Stop / transition: Recover earlier than planned if traffic, terrain, speed/load or weather is unfavorable. No real system disabling or descent toward an unsafe floor.

Reference: AFH: emergency descents; Private ACS: Emergency Descent; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 6, 8, & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2, 14, & 16

Current AIM: Chapters 1, 2, 4, & 9

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 42 - source page 63 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 43 - Airspace and communication review

Review airspace entry, airport/radio procedures and the specific solo airport/route authorizations before a nearby-airport trip.

[Return to lesson index](#)

Ground preparation objectives

- Review airspace entry, airport/radio procedures and the specific solo airport/route authorizations before a nearby-airport trip.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 2](#)

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 14-16](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapters 1-5 & 9](#)

[Sporty's course preparation: Vol 1: Segment 20](#)

[Sporty's course preparation: Vol 4: Segment 21](#)

[Sporty's course preparation: Vol 5: Segments 1, 2](#)

[Sporty's course preparation: Vol 6: Segments 5, 15](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 44 · Solo nearby-airport trip and authorized landing practice

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90 min solo | Brief 12 | Preflight 20 | Debrief 8

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-38 Air 08-30	[] Navigate by landmarks and magnetic compass	1.
38-54 Air 30-46	[] Arrive, land and depart at an unfamiliar airport	1.
54-66 Air 46-58	[] Short-field takeoff and landing	1.
66-76 Air 58-68	[] Soft-field takeoff and landing	1.
76-86 Air 68-78	[] Return, stabilized approach and final landing	1.
86-90 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 44

Fly to the specifically authorized airport within 25 NM and return. Practice approved landing techniques only if proficiency, runway and endorsement limits support them; handle actual uncertainty by seeking help.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Fly to the specifically authorized airport within 25 NM and return. Practice approved landing techniques only if proficiency, runway and endorsement limits support them; handle actual uncertainty by seeking help.
- Conduct only the specifically authorized route/maneuvers and return within the agreed limits; report actual repetitions and any difficulty.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 12 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Arrive, land and depart at an unfamiliar airport; Short-field takeoff and landing; Soft-field takeoff and landing.
- Before release, verify the student certificate/medical eligibility as applicable, current make/model solo endorsement, specific area/airport/route authority, weather/crosswind limits, fuel, aircraft status and contact/late-return plan. Solo is conditional on demonstrated proficiency; a calendar or timer never releases the student.

- Rehearse real lost/weather/equipment responses before departure. Do not create a lost condition, abnormal system state or unfamiliar maneuver alone; seek help or return when the authorization/safety margin changes.
- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Ground supervision: before release verify current solo authority, route/area, weather, fuel, aircraft condition and contact/return plan. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Ground supervision: observe the authorized departure if practical and retain the agreed contact plan. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 22 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Arrive, land and depart at an unfamiliar airport · 16 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Confirm destination position, runway/wind, published pattern and landing performance. 2. Establish radio awareness and visually identify traffic. 3. Join the correct pattern and fly a stable approach. 4. Land, clear and complete checks before heads-down taxi work. 5. Review departure/taxi routing before taking off again.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Finds the correct airport/runway, integrates safely with traffic and handles taxi/departure without instructor navigation rescue.

Watch / correct: Descending toward a guessed airport - remain safe and verify the airport before committing.

Watch / correct: Programming or folding charts while crossing a runway - stop at a safe location and perform the task there.

Stop / transition: Go around or divert if airport identity, runway suitability, traffic or weather is uncertain; obtain any required clearance before entering controlled movement/runway areas.

Reference: AIM: airport operations; current Chart Supplement, airport diagram and POH

Short-field takeoff and landing · 12 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 10 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Ground supervision: remain reachable, monitor the agreed return window and accept an early return/go-around without pressure to finish repetitions. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Ground supervision: meet/debrief after shutdown and verify the actual route, landing counts and any difficulty before another authorization. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.

- Student safely completes the specifically authorized mission and reports actual route, repetitions, full stops and any unplanned assistance/early return. The instructor verifies the required source mission rather than assuming it from flight time.

Debrief · 8 minutes

- Reconstruct the actual route/circuits, landing counts, weather/traffic and any deviation from the authorization.
- Identify any point where help, a go-around, return or diversion was needed; rehearse the response before another solo release.
- Confirm the airplane is secured and reconcile actual experience and next authorization with the instructor outside this app.

Study and sources

FAA Airplane Flying Handbook: Chapters 6 & 9

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 11 & 14

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 44 - source page 65 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 45 · Electronic navigation and automation preparation

Prepare installed VOR/GPS and autopilot capabilities, limitations and safe mode/position checks before later navigation training.

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Ground preparation objectives

- Prepare installed VOR/GPS and autopilot capabilities, limitations and safe mode/position checks before later navigation training.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Airplane Flying Handbook: Chapters 6, 9, & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16

Current AIM: Chapters 1 & 6

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course preparation: Vol 4: Segment 12,13

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 46 · Stage II preparation: performance, navigation and emergencies

[Return to lesson index](#)

60 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-24 Air 08-16	[] Short-field takeoff and landing	1.
24-32 Air 16-24	[] Soft-field takeoff and landing	1.
32-38 Air 24-30	[] Navigate by landmarks and magnetic compass	1.
38-46 Air 30-38	[] Choose and fly a diversion	1.
46-56 Air 38-48	[] Return, stabilized approach and final landing	1.
56-60 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 46

Assess current-ACS performance on field techniques and en-route decisions. The one-hour source target needs a compact safe local route and may not accommodate every task independently.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Assess current-ACS performance on field techniques and en-route decisions. The one-hour source target needs a compact safe local route and may not accommodate every task independently.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Finish required training and prepare an assessor plan of action. Identify prior deficiencies and unobserved tasks before the assessment.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Short-field takeoff and landing; Soft-field takeoff and landing; Navigate by landmarks and magnetic compass; Choose and fly a diversion.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Short-field takeoff and landing · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Navigate by landmarks and magnetic compass · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Choose and fly a diversion · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.

- Demonstrate applicable current Private ACS knowledge, risk management and skill without coaching. A missing task or safety intervention prevents a complete assessment; this guide grants no certificate or school examining authority.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Recover an uncertain position: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Recognize an equipment failure and choose a safe action: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 9 & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 14 & 16

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 46 - source page 67 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 47 · Stage II: navigation and field-performance assessment

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72 min check | Brief 48 | Preflight 15 | Debrief 24

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-26 Air 08-18	[] Short-field takeoff and landing	1.
26-36 Air 18-28	[] Soft-field takeoff and landing	1.
36-44 Air 28-36	[] Navigate by landmarks and magnetic compass	1.
44-52 Air 36-44	[] Choose and fly a diversion	1.
52-58 Air 44-50	[] Recover an uncertain position	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 47

Use an independent plan of action covering performance, pilotage, diversion/lost procedures and emergency descent/approach. Continue or reschedule unobserved tasks rather than issuing a clock-based pass.

Source clarification: The lesson page allows 72 discussion minutes; the allocation table lists 90. The visible source target follows the lesson page. Add discussion or another flight when the assessment scope needs it.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Use an independent plan of action covering performance, pilotage, diversion/lost procedures and emergency descent/approach. Continue or reschedule unobserved tasks rather than issuing a clock-based pass.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Finish required training and prepare an assessor plan of action. Identify prior deficiencies and unobserved tasks before the assessment.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 48 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Short-field takeoff and landing; Soft-field takeoff and landing; Navigate by landmarks and magnetic compass; Choose and fly a diversion; Recover an uncertain position.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current

ACS standards where applicable; Stage I uses the solo-readiness gate.

- Use the parked oral to assess certificates/endorsements, airworthiness, current weather, aircraft performance/systems, airspace and human factors. Include airport signs/markings/lighting and emergency/survival equipment where in scope. Final assessments also cover the planned cross-country, night preparation and knowledge-test deficiency codes.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Short-field takeoff and landing · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Navigate by landmarks and magnetic compass · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Choose and fly a diversion · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Recover an uncertain position · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Admit uncertainty promptly. 2. Maintain control and a safe altitude/course. 3. Check last known position, elapsed time and fuel. 4. Use appropriate navigation/ATC assistance to establish position. 5. Choose a safe landing plan rather than wandering.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace.

Watch / correct: Continuing random turns while searching - stabilize the airplane and work from the last known fix.

Watch / correct: Delaying a radio request through embarrassment - state uncertainty and request assistance early.

Stop / transition: Stop any simulation before actual uncertainty occurs; for a real lost condition, prioritize safety and obtain assistance. Never intentionally get lost on solo.

Reference: PHAK: lost procedures; AIM: assistance and communications

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Demonstrate applicable current Private ACS knowledge, risk management and skill without coaching. A missing task or safety intervention prevents a complete assessment; this guide grants no certificate or school examining authority.

Debrief · 24 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

Current Private Pilot Airplane ACS: knowledge, risk and skill review

Sporty's Academy Private TCO V20: lesson 47 - source page 69 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Cross-country, instruments, night and readiness

Day/solo cross-country, view-limited control, night missions and current-ACS mock/readiness assessment with explicit unfinished-task gates.

Lesson 48 · Instrument-control preparation

Study attitude/performance instruments and positive control transfer before view-limited VFR training.

[Return to lesson index](#)

Ground preparation objectives

- Study attitude/performance instruments and positive control transfer before view-limited VFR training.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

FAA Airplane Flying Handbook: Chapters 3 & 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 6 & 8

Current AIM: Chapters 1 & 6

Sporty's course preparation: Vol 5: Segments 17- 19

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 49 · Cross-country planning exercise

Bring a completed route, weather/NOTAM review, navigation log, fuel plan and alternate decisions to the day cross-country briefing.

[Return to lesson index](#)

Ground preparation objectives

- Bring a completed route, weather/NOTAM review, navigation log, fuel plan and alternate decisions to the day cross-country briefing.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 18](#)

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 2 & 9-17](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapters 1-9](#)

[Sporty's course: review the sections related to this flight](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 50 · Day cross-country: dead reckoning and 30-minute instrument introduction

[Return to lesson index](#)

90 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-30 Air 08-22	[] Dead reckoning and in-flight prediction	1.
30-52 Air 22-44	[] Basic instrument control under a view-limiting device	1.
52-60 Air 44-52	[] Instrument unusual-attitude recognition and recovery	1.
60-70 Air 52-62	[] VOR navigation and radar/ATC services when available	1.
70-76 Air 62-68	[] GPS route, nearest airport and safe automation	1.
76-86 Air 68-78	[] Return, stabilized approach and final landing	1.
86-90 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 50

Teach predictions from heading/time/wind and introduce instrument control during a VFR cross-country. Preserve the 30-minute instrument subset inside the 90-minute flight target.

Source clarification: Thirty instrument minutes overlap the flight target. VOR and autopilot exercises depend on installed, approved equipment; absent equipment is conditional, not an accomplished task.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Teach predictions from heading/time/wind and introduce instrument control during a VFR cross-country. Preserve the 30-minute instrument subset inside the 90-minute flight target.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Dead reckoning and in-flight prediction; Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery; VOR navigation and radar/ATC services when available; GPS route, nearest airport and safe automation.

- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Dead reckoning and in-flight prediction · 14 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show how the planned course, wind correction, groundspeed and leg time produce a predicted position. At the next visual checkpoint compare prediction with observation, then revise the remaining leg time and fuel estimate.

Student exercise: 1. Record departure/checkpoint time. 2. Fly the calculated heading. 3. Predict when and where the next checkpoint should appear. 4. Verify with pilotage and revise wind/groundspeed. 5. Recompute ETA and fuel for the next leg.

Practice: Two predicted checkpoints and two independent checks of the prediction.

Look for: Uses heading and elapsed time to predict position and notices a discrepancy before it becomes a lost-position problem.

Watch / correct: Recalculating heads-down for too long - write only essentials and pause calculation while controlling/looking outside.

Watch / correct: Accepting a predicted fix without visual confirmation - compare with two ground features.

Stop / transition: Suspend calculation for traffic, workload, worsening weather or position uncertainty; aviate and use an appropriate navigation/ATC safety resource.

Reference: PHAK: dead reckoning and wind correction; Private ACS: Navigation

Basic instrument control under a view-limiting device · 22 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 8 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

VOR navigation and radar/ATC services when available · 10 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: If equipped and supported by a usable signal, show tuning, identification, course selection and sensing. Demonstrate an intercept and tracking correction. Brief when radar/flight-following assistance helps and how to request it without surrendering VFR responsibility.

Student exercise: 1. Verify the facility and usable signal. 2. Set the desired course and interpret TO/FROM/sensing. 3. Intercept with a reasonable angle. 4. Track with wind correction and cross-check position visually. 5. Make the assigned radar-service request when appropriate.

Practice: One intercept and a tracked segment; one concise ATC service request or a parked rehearsal if service is unavailable.

Look for: Identifies and interprets the aid correctly, tracks without chasing the needle and keeps VFR terrain/weather responsibility.

Watch / correct: Tracking without checking identification - identify the signal before using it.

Watch / correct: Turning the wrong way from reverse sensing - recheck course and TO/FROM before intercepting.

Stop / transition: Stop the navigation exercise for unreliable signal, unsafe course, weather/terrain conflict or excessive heads-down time. Mark unavailable equipment as not applicable, not as demonstrated.

Reference: PHAK: navigation systems; AIM: VOR and radar services

GPS route, nearest airport and safe automation · 6 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Show the installed unit's approved operating method for checking the active route, sequencing and nearest-airport page. Compare database information with current charts and airport suitability. If an autopilot is installed, demonstrate mode/annunciation checks and immediate manual override using its supplement.

Student exercise: 1. Verify the active waypoint/route against the chart. 2. Confirm mode and sequencing. 3. Use nearest to find options, then check runway/weather/fuel suitability. 4. If approved and equipped, engage a briefed autopilot mode, monitor it and disengage manually. 5. Keep a visual position cross-check.

Practice: One route verification, one nearest-airport choice and one monitored automation engage/disengage cycle if equipped.

Look for: Understands what the installed unit is commanding, chooses a suitable option rather than merely the closest airport and can return promptly to manual control.

Watch / correct: Following the magenta line without checking the waypoint - verify the active leg against the planned course.

Watch / correct: Assuming an armed mode is active - read the actual annunciation and monitor the flight path.

Stop / transition: Disengage automation and fly manually for unexpected behavior, mode uncertainty or unsafe path; discontinue programming for high workload. Use only aircraft-approved equipment procedures.

Reference: Installed avionics/autopilot supplement; PHAK: navigation; Private ACS: Navigation Systems

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.

Debrief - 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Study and sources

FAA Airplane Flying Handbook: Chapters 3 & 5

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 6 & 8

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 50 - source page 73 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 51 · Day cross-country: GPS, tower operations and instrument review

[Return to lesson index](#)

90 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-23 Air 08-15	[] Navigate by landmarks and magnetic compass	1.
23-45 Air 15-37	[] Basic instrument control under a view-limiting device	1.
45-53 Air 37-45	[] Instrument unusual-attitude recognition and recovery	1.
53-63 Air 45-55	[] GPS route, nearest airport and safe automation	1.
63-76 Air 55-68	[] Towered-airport arrival and full-stop operation	1.
76-86 Air 68-78	[] Return, stabilized approach and final landing	1.
86-90 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 51

Use GPS route/nearest functions, a towered airport and instrument review while continuing visual navigation. Demonstrate ADF/NDB and autopilot only when installed and usable.

Source clarification: The 90-minute target includes 30 instrument minutes. A towered arrival and safe route may require longer; VOR/ADF/autopilot practice can be carried forward when equipment, facility or time is unsuitable.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Use GPS route/nearest functions, a towered airport and instrument review while continuing visual navigation. Demonstrate ADF/NDB and autopilot only when installed and usable.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery; GPS route, nearest airport and safe automation; Towered-airport arrival and full-stop operation.

- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 7 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Show the next prominent checkpoint, chart-to-ground orientation and a heading checked with the compass in stable flight. Compare planned and actual time/fuel. Demonstrate a concise position report or Flight Service contact when useful, without letting communications replace lookout.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: At least three checkpoint fixes and two time/fuel updates; use the source route distances rather than a generic local circuit.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Basic instrument control under a view-limiting device · 22 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 8 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

GPS route, nearest airport and safe automation · 10 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Show the installed unit's approved operating method for checking the active route, sequencing and nearest-airport page. Compare database information with current charts and airport suitability. If an autopilot is installed, demonstrate mode/annunciation checks and immediate manual override using its supplement.

Student exercise: 1. Verify the active waypoint/route against the chart. 2. Confirm mode and sequencing. 3. Use nearest to find options, then check runway/weather/fuel suitability. 4. If approved and equipped, engage a briefed autopilot mode, monitor it and disengage manually. 5. Keep a visual position cross-check.

Practice: One route verification, one nearest-airport choice and one monitored automation engage/disengage cycle if equipped.

Look for: Understands what the installed unit is commanding, chooses a suitable option rather than merely the closest airport and can return promptly to manual control.

Watch / correct: Following the magenta line without checking the waypoint - verify the active leg against the planned course.

Watch / correct: Assuming an armed mode is active - read the actual annunciation and monitor the flight path.

Stop / transition: Disengage automation and fly manually for unexpected behavior, mode uncertainty or unsafe path; discontinue programming for high workload. Use only aircraft-approved equipment procedures.

Reference: Installed avionics/autopilot supplement; PHAK: navigation; Private ACS: Navigation Systems

Towered-airport arrival and full-stop operation · 13 minutes

Teach / coach: Ask the student to state the setup and attempt the exercise first. Demonstrate a missed element after a safe reset, then return control: Brief the airport class, frequencies, entry requirements, runway and diagram. Model one concise call when needed. Explain clearance versus acknowledgment, hold-short discipline and light-signal actions before arrival.

Student exercise: 1. Establish required communication/clearance before entry. 2. Read back assigned runway/hold-short instructions. 3. Integrate with traffic and land only with the required clearance. 4. Clear the runway and change frequency as directed. 5. Taxi using the diagram and stop if any instruction is uncertain.

Practice: One towered arrival-full-stop-departure sequence; source-required multiple full stops must be scheduled explicitly, not inferred from one circuit.

Look for: Meets entry requirements, understands the runway clearance and taxis without crossing a hold line on an uncertain instruction.

Watch / correct: Hearing a call sign as a landing clearance - identify the actual clearance before continuing.

Watch / correct: Guessing an unfamiliar taxi route - stop safely and ask for clarification/progressive taxi.

Stop / transition: Hold, go around or request assistance if clearance, runway identity or traffic is uncertain; never enter prohibited airspace or cross a runway to preserve the timetable.

Reference: AIM: ATC communications, light signals and airport operations; current airport diagram

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- VOR navigation and radar/ATC services when available: conditional on installed equipment and a usable facility; not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- ADF/NDB homing when installed and usable: conditional on installed equipment and a usable facility; not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring

the source lesson complete.

Study and sources

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16

Current AIM: Chapters 1-5

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 51 - source page 74 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 52 · Solo cross-country: 150 NM, three airports and tower full stops

[Return to lesson index](#)

120 min solo | Brief 12 | Preflight 20 | Debrief 8

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-41 Air 08-33	[] Navigate by landmarks and magnetic compass	1.
41-66 Air 33-58	[] Towered-airport full-stop takeoff and landing work	1.
66-86 Air 58-78	[] Navigate by landmarks and magnetic compass	1.
86-106 Air 78-98	[] Arrive, land and depart at an unfamiliar airport	1.
106-116 Air 98-108	[] Return, stabilized approach and final landing	1.
116-120 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 52

Complete the authorized three-leg solo route with three landing points, at least 150 NM total and one straight-line segment longer than 50 NM. Also schedule the source-required towered full-stop work; verify actual counts and distance.

Source clarification: The 120-minute publisher target may not fit the route and all required full stops. Plan actual leg/taxi times and fuel; extend or schedule additional authorized flying instead of reducing the 150 NM, three landing points, >50 NM leg or required takeoff/landing counts.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Complete the authorized three-leg solo route with three landing points, at least 150 NM total and one straight-line segment longer than 50 NM. Also schedule the source-required towered full-stop work; verify actual counts and distance.
- Conduct only the specifically authorized route/maneuvers and return within the agreed limits; report actual repetitions and any difficulty.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- The authorized instructor must verify required logged training, proficiency, knowledge-test corrections and applicable endorsements before any solo release.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 12 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.

- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Towered-airport arrival and full-stop operation; Navigate by landmarks and magnetic compass; Arrive, land and depart at an unfamiliar airport.
- Before release, verify the student certificate/medical eligibility as applicable, current make/model solo endorsement, specific area/airport/route authority, weather/crosswind limits, fuel, aircraft status and contact/late-return plan. Solo is conditional on demonstrated proficiency; a calendar or timer never releases the student.
- Rehearse real lost/weather/equipment responses before departure. Do not create a lost condition, abnormal system state or unfamiliar maneuver alone; seek help or return when the authorization/safety margin changes.
- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Ground supervision: before release verify current solo authority, route/area, weather, fuel, aircraft condition and contact/return plan. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Ground supervision: observe the authorized departure if practical and retain the agreed contact plan. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 25 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Towered-airport full-stop takeoff and landing work · 25 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Establish required communication/clearance before entry. 2. Read back assigned runway/hold-short instructions. 3. Integrate with traffic and land only with the required clearance. 4. Clear the runway and change frequency as directed. 5. Taxi using the diagram and stop if any instruction is uncertain. Complete the three source-required takeoffs and full-stop landings at an airport with an operating control tower, using a traffic pattern for each. Ask for taxi/pattern instructions and count each actual full stop; extend or add authorized flying if the 25-minute block does not safely fit all three.

Practice: Three actual towered full-stop takeoffs/landings with a traffic pattern for each; the target block must be extended/continued if actual traffic/taxi time prevents completion.

Look for: Meets entry requirements, understands the runway clearance and taxis without crossing a hold line on an uncertain instruction.

Watch / correct: Hearing a call sign as a landing clearance - identify the actual clearance before continuing.

Watch / correct: Guessing an unfamiliar taxi route - stop safely and ask for clarification/progressive taxi.

Stop / transition: Hold, go around or request assistance if clearance, runway identity or traffic is uncertain; never enter prohibited airspace or cross a runway to preserve the timetable.

Reference: AIM: ATC communications, light signals and airport operations; current airport diagram

Navigate by landmarks and magnetic compass · 20 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Arrive, land and depart at an unfamiliar airport · 20 minutes

Teach / coach: Before departure, verify prior dual demonstration and proficiency, then authorize this exact task, location, repetitions and weather limits. Review the student's setup and stop criteria using the aircraft references. Remain available under the agreed supervision/contact plan. No in-flight instructor demonstration or new-task training is possible; missing proficiency means another dual lesson before release.

Student exercise: 1. Confirm destination position, runway/wind, published pattern and landing performance. 2. Establish radio awareness and visually identify traffic. 3. Join the correct pattern and fly a stable approach. 4. Land, clear and complete checks before heads-down taxi work. 5. Review departure/taxi routing before taking off again.

Practice: Perform only the prebriefed authorized repetitions within the allotted block. Skip a task rather than exceed weather, proficiency, endorsement or fuel limits.

Look for: Finds the correct airport/runway, integrates safely with traffic and handles taxi/departure without instructor navigation rescue.

Watch / correct: Descending toward a guessed airport - remain safe and verify the airport before committing.

Watch / correct: Programming or folding charts while crossing a runway - stop at a safe location and perform the task there.

Stop / transition: Go around or divert if airport identity, runway suitability, traffic or weather is uncertain; obtain any required clearance before entering controlled movement/runway areas.

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Ground supervision: remain reachable, monitor the agreed return window and accept an early return/go-around without pressure to finish repetitions. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Ground supervision: meet/debrief after shutdown and verify the actual route, landing counts and any difficulty before another authorization. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.

- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student safely completes the specifically authorized mission and reports actual route, repetitions, full stops and any unplanned assistance/early return. The instructor verifies the required source mission rather than assuming it from flight time.
- Verify at least 150 NM total, three landing points, one straight-line leg longer than 50 NM, source-required towered full-stop work, all route endorsements and actual landing counts; use current Part 61 experience rules when reconciling the complete solo block.

Debrief - 8 minutes

- Reconstruct the actual route/circuits, landing counts, weather/traffic and any deviation from the authorization.
- Identify any point where help, a go-around, return or diversion was needed; rehearse the response before another solo release.
- Confirm the airplane is secured and reconcile actual experience and next authorization with the instructor outside this app.

Study and sources

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 16

Current AIM: Chapters 1-5

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 52 - source page 75 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 53 - Night preparation

Prepare night lighting, visual illusions, personal readiness, terrain risks and night regulatory definitions before the first night flight.

[Return to lesson index](#)

Ground preparation objectives

- Prepare night lighting, visual illusions, personal readiness, terrain risks and night regulatory definitions before the first night flight.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[FAA Airplane Flying Handbook: Chapter 11](#)

[FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 17](#)

[Current Part 61 and applicable Part 91 rules](#)

[Current AIM: Chapters 2, 4, & 7](#)

[Sporty's course preparation: Vol 4: Segments 1-2](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 54 · First night flight with 30-minute instrument review

[Return to lesson index](#)

60 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-38 Air 08-30	[] Basic instrument control under a view-limiting device	1.
38-46 Air 30-38	[] Instrument unusual-attitude recognition and recovery	1.
46-56 Air 38-48	[] Return, stabilized approach and final landing	1.
56-60 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 54

Introduce lighting, illusions and night landing judgment, plus the source 30-minute instrument subset. Five full-stop night landings are a source goal; the one-hour budget is often too tight for both.

Source clarification: Thirty instrument minutes plus taxi/transit leave no useful repeated-pattern budget in 60 minutes. This target flight introduces one night arrival; extend or schedule another dual session to complete the five full-stop takeoffs/landings, night go-around and emergency work. Do not mark the source lesson complete until the missing work is finished.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Introduce lighting, illusions and night landing judgment, plus the source 30-minute instrument subset. Five full-stop night landings are a source goal; the one-hour budget is often too tight for both.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery.

- Check night medical/personal readiness, dark adaptation, cockpit/external/backup lighting, runway and terrain references. Discuss illusions, lighting failures and suitable landing options. Count actual full-stop takeoffs/landings and use the applicable regulatory night definitions for the purpose being logged.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry. Verify usable external/cockpit lights and the backup flashlight before moving into darkness.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Basic instrument control under a view-limiting device • 22 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery • 8 minutes

Teach / coach: Demonstrate with an explicit control transfer, then coach the student: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early. Positively identify night runway lights and use the approved path references; do not descend toward unverified darkness.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.

- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student explains the new technique and performs it with the assistance appropriate to an introduction; new-task success is not final ACS mastery.
- Complete the source five full-stop night takeoffs/landings, night go-around and emergency work in an extended/additional dual flight if they do not fit this target. Verify actual 30-minute instrument practice rather than assuming the timer is logged instrument time.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Night orientation, lighting and landing judgment: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Night emergency options and a night go-around: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 11 & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 17

Current Part 61 and applicable Part 91 rules

Current AIM: Chapters 2, 4, & 7

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 54 - source page 77 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 55 · Night cross-country, full-stop landings and instrument review

[Return to lesson index](#)

120 min dual | Brief 8 | Preflight 20 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-36 Air 08-28	[] Navigate by landmarks and magnetic compass	1.
36-58 Air 28-50	[] Basic instrument control under a view-limiting device	1.
58-66 Air 50-58	[] Instrument unusual-attitude recognition and recovery	1.
66-76 Air 58-68	[] GPS route, nearest airport and safe automation	1.
76-96 Air 68-88	[] Night orientation, lighting and landing judgment	1.
96-106 Air 88-98	[] Night emergency options and a night go-around	1.
106-116 Air 98-108	[] Return, stabilized approach and final landing	1.
116-120 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 55

Plan a night trip over 100 NM total with night navigation, 30 instrument minutes and at least five full-stop takeoffs/landings. Reconcile actual night experience before considering the night block complete.

Source clarification: The 120-minute target is not a promise that >100 NM and five full-stop cycles fit. Use real aircraft/wind/taxi estimates and add time or another dual flight. Verify the Part 61 night requirement, cumulative three night hours and ten night full-stop takeoffs/landings, and the cross-country distance; instrument time overlaps this flight.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Plan a night trip over 100 NM total with night navigation, 30 instrument minutes and at least five full-stop takeoffs/landings. Reconcile actual night experience before considering the night block complete.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 20 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery; GPS route, nearest airport and safe automation; Night orientation, lighting and landing judgment; Night emergency options and a

night go-around.

- Bring current charts, Chart Supplement/airport diagrams, a weather/NOTAM briefing, route/heading/time/fuel log and alternatives. Check each source-required distance/airport/landing against the actual route; calculated flight duration and reserves decide the booking length.
- Check night medical/personal readiness, dark adaptation, cockpit/external/backup lighting, runway and terrain references. Discuss illusions, lighting failures and suitable landing options. Count actual full-stop takeoffs/landings and use the applicable regulatory night definitions for the purpose being logged.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry. Verify usable external/cockpit lights and the backup flashlight before moving into darkness.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 20 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show the next prominent checkpoint, chart-to-ground orientation and a heading checked with the compass in stable flight. Compare planned and actual time/fuel. Demonstrate a concise position report or Flight Service contact when useful, without letting communications replace lookout.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: At least three checkpoint fixes and two time/fuel updates; use the source route distances rather than a generic local circuit.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Basic instrument control under a view-limiting device · 22 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

GPS route, nearest airport and safe automation · 10 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show the installed unit's approved operating method for checking the active route, sequencing and nearest-airport page. Compare database information with current charts and airport suitability. If an autopilot is installed, demonstrate mode/annunciation checks and immediate manual override using its supplement.

Student exercise: 1. Verify the active waypoint/route against the chart. 2. Confirm mode and sequencing. 3. Use nearest to find options, then check runway/weather/fuel suitability. 4. If approved and equipped, engage a briefed autopilot mode, monitor it and disengage manually. 5. Keep a visual position cross-check.

Practice: One route verification, one nearest-airport choice and one monitored automation engage/disengage cycle if equipped.

Look for: Understands what the installed unit is commanding, chooses a suitable option rather than merely the closest airport and can return promptly to manual control.

Watch / correct: Following the magenta line without checking the waypoint - verify the active leg against the planned course.

Watch / correct: Assuming an armed mode is active - read the actual annunciation and monitor the flight path.

Stop / transition: Disengage automation and fly manually for unexpected behavior, mode uncertainty or unsafe path; discontinue programming for high workload. Use only aircraft-approved equipment procedures.

Reference: Installed avionics/autopilot supplement; PHAK: navigation; Private ACS: Navigation Systems

Night orientation, lighting and landing judgment · 20 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show the dimmed cockpit, outside scan and identification of runway/obstruction lighting. Demonstrate a night pattern with the approved airspeed/descent path and explain black-hole, false-horizon and distance illusions; keep an approved visual glide reference when available.

Student exercise: 1. Verify lights, backup illumination and night readiness. 2. Identify airport/runway lights positively. 3. Fly the briefed pattern and stabilized approach. 4. Maintain centerline and the approved landing picture. 5. Complete a full stop and check lighting before another departure.

Practice: A demonstrated circuit on introduction followed by student full-stop takeoffs/landings; record actual full-stop counts, not touch-and-go equivalents.

Look for: Identifies the correct runway, manages lighting/illusions and maintains a stable approach without descending toward unverified dark terrain.

Watch / correct: Judging glide path only by runway appearance - use the approved visual/flight-path references and cross-check.

Watch / correct: Brightening the cockpit excessively - restore readable dim lighting and preserve outside adaptation.

Stop / transition: Go around or divert for uncertain runway identity, lighting failure, visual illusion, unstable path or weather/terrain uncertainty; never descend into darkness merely to regain a runway sight picture.

Reference: AFH: night operations; PHAK: human factors; AIM: airport lighting; POH

Night emergency options and a night go-around · 10 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Before flight, identify reachable lighted landing options and the POH response to a lighting/electrical failure. In flight use verbal scenarios while preserving real illumination; demonstrate or assess a safe night go-around without manufacturing a hazardous approach.

Student exercise: 1. Keep aircraft control during the announced scenario. 2. Select the correct checklist and verify navigation/lighting options. 3. Choose a suitable landing/diversion plan. 4. Fly one briefed night go-around with approved configuration changes. 5. Explain the terrain and visual-illusion risks.

Practice: One verbal equipment scenario and one night go-around; no intentional loss of essential lights or unexpected power reduction near the ground.

Look for: Prioritizes control, chooses a usable landing option and carries out the go-around while maintaining runway/terrain clearance.

Watch / correct: Looking for an unlighted field without a safe path - use known suitable options and preserve terrain clearance.

Watch / correct: Reaching for a flashlight during a critical climb - fly first and use the agreed backup-light location when workload allows.

Stop / transition: Stop the simulation for any real light/electrical fault or loss of visual reference; instructor intervenes before terrain or aircraft control is threatened.

Reference: AFH: night emergencies and go-arounds; POH: electrical and emergency procedures

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early. Positively identify night runway lights and use the approved path references; do not descend toward unverified darkness.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.

- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.
- Verify the night cross-country exceeds 100 NM total and this lesson's five full-stop cycles; reconcile cumulative night hours/landings and all applicable Part 61 requirements before the night block is complete.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- VOR navigation and radar/ATC services when available: conditional on installed equipment and a usable facility; not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- ADF/NDB homing when installed and usable: conditional on installed equipment and a usable facility; not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 11 & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 16 & 17

Current AIM: Chapters 1-5 & 7

Current Private Pilot Airplane ACS: lesson-related tasks

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 55 - source page 78 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 56 · Stage III review: navigation, performance and instrument control

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90 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-38 Air 08-30	[] Basic instrument control under a view-limiting device	1.
38-46 Air 30-38	[] Instrument unusual-attitude recognition and recovery	1.
46-56 Air 38-48	[] Short-field takeoff and landing	1.
56-66 Air 48-58	[] Soft-field takeoff and landing	1.
66-76 Air 58-68	[] Choose and fly a diversion	1.
76-86 Air 68-78	[] Return, stabilized approach and final landing	1.
86-90 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 56

Review oral decisions and Stage II flight skills, plus the source 30-minute instrument target. Select the weakest areas first and record all unobserved source tasks for additional training.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Review oral decisions and Stage II flight skills, plus the source 30-minute instrument target. Select the weakest areas first and record all unobserved source tasks for additional training.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery; Short-field takeoff and landing; Soft-field takeoff and landing; Choose and fly a diversion.
- Use the parked oral to assess certificates/endorsements, airworthiness, current weather, aircraft performance/systems, airspace and human factors. Include airport signs/markings/lighting and emergency/survival equipment where in scope. Final assessments also cover the planned cross-country,

night preparation and knowledge-test deficiency codes.

- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Basic instrument control under a view-limiting device · 22 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

Short-field takeoff and landing · 10 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Use POH performance, actual runway conditions and a selected obstacle/aiming point. Demonstrate the approved takeoff setup, obstacle-clearance climb and transition. On return show stabilized approach speed, precise touchdown and approved stopping technique.

Student exercise: 1. Calculate usable runway and margins. 2. Apply the POH short-field takeoff procedure. 3. Track centerline and transition only after obstacle clearance. 4. Fly a stable short-field approach at the approved speed. 5. Touch down at the selected point within the applicable lesson/ACS tolerance and stop with directional control.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review if the runway/pattern supports them.

Look for: Uses performance data and approved procedures, clears the simulated obstacle and controls touchdown/stopping without chasing a spot unsafely.

Watch / correct: Rotating early to improve distance - use the POH procedure and required control speed.

Watch / correct: Pitching down to hit the spot - go around when the approach cannot remain stable.

Stop / transition: Reject or go around for performance margin, unstable speed, obstacle-clearance uncertainty or poor alignment. Simulate a field condition on a suitable runway rather than using an unsafe short strip.

Reference: AFH: short-field takeoffs and landings; Private ACS: short-field tasks; POH

Soft-field takeoff and landing · 10 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Explain how to unload the nosewheel and preserve momentum using the POH technique. Demonstrate the ground-effect transition without an early climb at insufficient airspeed. On landing show energy management and protection of the landing gear.

Student exercise: 1. Brief the approved configuration and simulated surface. 2. Maintain the required nosewheel unloading and directional control during the takeoff roll. 3. Accelerate safely in ground effect as prescribed. 4. Transition to the approved climb. 5. Fly a stable soft-field approach and protect the gear through touchdown and roll.

Practice: One demonstration and one student takeoff-landing pair on introduction; two student pairs on review when safe.

Look for: Maintains control and momentum, avoids premature climb and manages touchdown without dropping the nosewheel.

Watch / correct: Pulling out of ground effect before acceleration - lower the pitch appropriately and follow the POH transition.

Watch / correct: Trying to hold the nose up beyond control margin - prioritize directional control and aircraft limits.

Stop / transition: Stop the simulation for inadequate surface/runway suitability or go around for an unstable approach. Use actual soft surfaces only when approved and suitable.

Reference: AFH: soft-field takeoffs and landings; Private ACS: soft-field tasks; POH

Choose and fly a diversion · 10 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Present a briefed weather, fuel or destination problem without creating a real hazard. Show selecting a suitable alternative, estimating heading/time/fuel, checking terrain/airspace and changing the flight plan or informing ATC as appropriate.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One complete diversion decision and two position/ETA updates along the new course.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Navigate by landmarks and magnetic compass: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Recover an uncertain position: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 11 & 18

FAA Pilot's Handbook of Aeronautical Knowledge: Chapter 17

Current Part 61 and applicable Part 91 rules

Current AIM: Chapters 2, 4, & 7

Sporty's course: review the sections related to this flight

Sporty's Academy Private TCO V20: lesson 56 - source page 79 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 57 · Knowledge test and stage review

Review knowledge-test deficiency codes, current ACS subjects and experience/endorsement readiness before the mock practical test.

[Return to lesson index](#)

Ground preparation objectives

- Review knowledge-test deficiency codes, current ACS subjects and experience/endorsement readiness before the mock practical test.

Preparation

- Complete the corresponding current Sporty's ground lesson and its assigned preparation before dependent flight training.

[Current Private Pilot Airplane ACS: knowledge, risk and skill review](#)

Complete the corresponding ground lesson and bring questions to the flight briefing. This page is a preparation pointer; the flight teaching guides begin at flight lessons.

Lesson 57a · End-of-stage mock practical test

[Return to lesson index](#)

120 min check | Brief 80 | Preflight 15 | Debrief 40

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-28 Air 08-20	[] Navigate by landmarks and magnetic compass	1.
28-38 Air 20-30	[] Choose and fly a diversion	1.
38-44 Air 30-36	[] Recover an uncertain position	1.
44-52 Air 36-44	[] Steep turns in both directions	1.
52-58 Air 44-50	[] Rectangular course: wind correction by leg	1.
58-64 Air 50-56	[] S-turns across a road	1.
64-70 Air 56-62	[] Turns around a point	1.
70-76 Air 62-68	[] Slow flight: configuration, coordination and exit	1.
76-82 Air 68-74	[] Full power-off stalls: approach scenario	1.
82-88 Air 74-80	[] Full power-on stalls: departure scenario	1.
88-100 Air 80-92	[] Basic instrument control under a view-limiting device	1.
100-106 Air 92-98	[] Instrument unusual-attitude recognition and recovery	1.
106-116 Air 98-108	[] Return, stabilized approach and final landing	1.
116-120 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 57a

Use a different qualified instructor where practicable. Assess current Private ACS knowledge, risk management and skills with no training prompts, including an 18-minute instrument segment.

Source clarification: The publisher includes 120 minutes of discussion and 120 flight minutes, with 18 instrument minutes inside the flight. Complete all required tasks using a safe plan of action; the target can require extension or a continued assessment rather than an incomplete-task pass.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Use a different qualified instructor where practicable. Assess current Private ACS knowledge, risk management and skills with no training prompts, including an 18-minute instrument segment.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Finish required training and prepare an assessor plan of action. Identify prior deficiencies and unobserved tasks before the assessment.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight • 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMS, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing • 80 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.

- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Choose and fly a diversion; Recover an uncertain position; Steep turns in both directions; Rectangular course: wind correction by leg; S-turns across a road; Turns around a point; Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario; Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.
- Use the parked oral to assess certificates/endorsements, airworthiness, current weather, aircraft performance/systems, airspace and human factors. Include airport signs/markings/lighting and emergency/survival equipment where in scope. Final assessments also cover the planned cross-country, night preparation and knowledge-test deficiency codes.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 12 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Choose and fly a diversion · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Recover an uncertain position · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Admit uncertainty promptly. 2. Maintain control and a safe altitude/course. 3. Check last known position, elapsed time and fuel. 4. Use appropriate navigation/ATC assistance to establish position. 5. Choose a safe landing plan rather than wandering.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace.

Watch / correct: Continuing random turns while searching - stabilize the airplane and work from the last known fix.

Watch / correct: Delaying a radio request through embarrassment - state uncertainty and request assistance early.

Stop / transition: Stop any simulation before actual uncertainty occurs; for a real lost condition, prioritize safety and obtain assistance. Never intentionally get lost on solo.

Reference: PHAK: lost procedures; AIM: assistance and communications

Steep turns in both directions · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Rectangular course: wind correction by leg · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and establish the briefed entry. 2. Hold parallel ground tracks with wind correction. 3. Change bank smoothly through each corner to preserve spacing. 4. Call the wind correction on each leg. 5. Exit without descending or tightening toward obstacles.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn.

Watch / correct: Turning too late on the fast downwind corner - anticipate the corner and use an appropriate coordinated bank.

Watch / correct: Looking continuously at the field - alternate the reference with traffic, horizon and altitude scans.

Stop / transition: Exit toward a safe area for traffic, inadequate obstacle clearance, excessive bank or an unstable altitude trend.

Reference: AFH: rectangular course; Private ACS: Ground Reference Maneuvers

S-turns across a road · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and approach the reference as briefed. 2. Begin a coordinated turn that compensates for the wind. 3. Cross wings level and reverse the turn. 4. Adjust bank to create the second comparable arc. 5. Exit and describe the strongest drift correction.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination.

Watch / correct: Holding one bank all the way around - anticipate changing groundspeed and vary bank.

Watch / correct: Rushing the reversal with rudder - level the wings at the reference, then enter the opposite bank together.

Stop / transition: Exit if bank/altitude trend is unsafe or the chosen road has traffic/obstructions that compromise the practice area; maintain legal clearances.

Turns around a point · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and choose a safe radius and altitude. 2. Enter as briefed. 3. Adjust bank to maintain the ground radius. 4. Keep the ball centered and scan outside. 5. Exit after a full circuit and predict corrections for a second direction.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan.

Watch / correct: Tightening toward the point when blown inward - reduce bank and re-establish the planned radius.

Watch / correct: Fixating on the point - scan the horizon and traffic, then check the ground reference briefly.

Stop / transition: Leave the maneuver if the required bank exceeds the practice limit, an escape route is compromised or traffic/terrain clearance becomes uncertain.

Reference: AFH: turns around a point; Private ACS: Ground Reference Maneuvers

Slow flight: configuration, coordination and exit · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Basic instrument control under a view-limiting device • 12 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery • 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.

- Demonstrate applicable current Private ACS knowledge, risk management and skill without coaching. A missing task or safety intervention prevents a complete assessment; this guide grants no certificate or school examining authority.
- Confirm experience, tests, endorsements and the required preparation within the applicable regulatory period separately from this mock/readiness result. Review and correct knowledge-test deficiency areas before recommendation.

Debrief - 40 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Short-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Soft-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- VOR navigation and radar/ATC services when available: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

Current Private Pilot Airplane ACS: knowledge, risk and skill review

Sporty's Academy Private TCO V20: lesson 57a - source page 81 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 58 · Final remedial flight and instrument experience check

[Return to lesson index](#)

72 min dual | Brief 8 | Preflight 15 | Debrief 4

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-38 Air 08-30	[] Basic instrument control under a view-limiting device	1.
38-46 Air 30-38	[] Instrument unusual-attitude recognition and recovery	1.
46-52 Air 38-44	[] Steep turns in both directions	1.
52-58 Air 44-50	[] Slow flight: configuration, coordination and exit	1.
58-68 Air 50-60	[] Return, stabilized approach and final landing	1.
68-72 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 58

Use the previous assessment to choose targeted corrections, then reconcile actual instrument experience. The source broad task list cannot receive adequate repetitions inside 72 minutes with 30 minutes under the view limiter.

Source clarification: Choose the highest-priority 12-minute visual remediation block instead of assuming this entire source list is mastered. The shown default is steep turns plus slow flight. Extend/repeat to remedy any other deficiency and verify at least three actual instrument-training hours for the Part 61 airplane requirement.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Use the previous assessment to choose targeted corrections, then reconcile actual instrument experience. The source broad task list cannot receive adequate repetitions inside 72 minutes with 30 minutes under the view limiter.
- Explain the setup, fly the assigned exercise, diagnose the largest error and demonstrate a corrected repeat when the budget safely permits.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight · 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMs, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing · 8 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.
- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.

- Preview today's ordered flight blocks: Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery; Steep turns in both directions; Slow flight: configuration, coordination and exit.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Have the student lead the start/taxi checks; coach only a diagnosed error after stopping safely. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Have the student brief and fly the departure, diagnosing any drift/airspeed issue after a safe climb. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Basic instrument control under a view-limiting device · 22 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Establish instructor lookout and a positive control transfer before applying the view limiter. Demonstrate attitude-led straight flight, heading changes, constant-airspeed climbs/descents and level-offs. Explain a short cross-check and small corrections; this is VFR training with a safety lookout.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: Two heading changes each way plus one climb and one descent with level-offs; count only actual view-limited instrument time.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 8 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Agree on control transfers and modest, aircraft-approved setup boundaries. Demonstrate recognition of a nose-high or nose-low instrument picture and the POH/FAA recovery priorities. The instructor establishes the attitude and maintains traffic/terrain clearance.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One demonstrated and one student recovery for each attitude on introduction; one student recovery each way on review/check.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

Steep turns in both directions • 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Show clearing, the current Private ACS bank target, the added back pressure/power as needed and an outside reference that reveals altitude drift. Demonstrate the rollout and a complete reset before changing direction.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: One demonstration followed by one student turn each way on introduction; two student turns each way on review if safe and unrushed.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Slow flight: configuration, coordination and exit • 6 minutes

Teach / coach: Observe the first attempt without continuous prompting. After a safe reset, give one diagnosis, show the correction if needed and require another attempt: Clear the area, identify configuration limits and explain the selected slow-flight condition. Reduce power while maintaining altitude, configure in the POH sequence and coordinate added power. Show gentle turns and a deliberate return to cruise.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One entry, two shallow turns and one recovery; repeat the entry only after identifying the largest error.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Return, stabilized approach and final landing • 10 minutes

Teach / coach: Let the student brief and fly the arrival; diagnose one energy or alignment issue only after a safe landing/reset. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Student leads the setup and control with decreasing assistance, recognizes deviations and shows a corrected repeat. Final-stage proficiency is judged against the current Private ACS, using the actual aircraft references.
- Reconcile at least three actual hours of instrument training under the Part 61 single-engine-airplane experience requirement, plus any unresolved assessment task. Do not add overlapping instrument time again

to flight totals.

Debrief · 4 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Short-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Soft-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Rectangular course: wind correction by leg: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- S-turns across a road: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Turns around a point: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Full power-off stalls: approach scenario: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Full power-on stalls: departure scenario: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Navigate by landmarks and magnetic compass: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Choose and fly a diversion: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Recover an uncertain position: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

- VOR navigation and radar/ATC services when available: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

FAA Airplane Flying Handbook: Chapters 1-11 & 18

Current Private Pilot Airplane ACS: lesson-related tasks

FAA Pilot's Handbook of Aeronautical Knowledge: Chapters 1-17

Current AIM: Chapters 1-9

Sporty's Academy Private TCO V20: lesson 58 - source page 82 - Sporty's Academy Private 141 TCO V20 / 06-17-2026

Lesson 59 · Final Part 61 readiness assessment

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120 min check | Brief 80 | Preflight 15 | Debrief 40

Publisher lesson-page target; phase allocations authored by SUAREZ.CFI

Minutes are pacing targets. Stop, repeat, or carry forward when safety or proficiency requires. Set aircraft values from POH/AFM.

Cockpit sequence

Flight / air	Exercise	Action
00-08 Ground	[] Start, taxi, run-up and runway confirmation	1.
08-16 Air 00-08	[] Takeoff, climb and safe transit	1.
16-28 Air 08-20	[] Navigate by landmarks and magnetic compass	1.
28-38 Air 20-30	[] Choose and fly a diversion	1.
38-44 Air 30-36	[] Recover an uncertain position	1.
44-52 Air 36-44	[] Steep turns in both directions	1.
52-58 Air 44-50	[] Rectangular course: wind correction by leg	1.
58-64 Air 50-56	[] S-turns across a road	1.
64-70 Air 56-62	[] Turns around a point	1.
70-76 Air 62-68	[] Slow flight: configuration, coordination and exit	1.
76-82 Air 68-74	[] Full power-off stalls: approach scenario	1.
82-88 Air 74-80	[] Full power-on stalls: departure scenario	1.
88-100 Air 80-92	[] Basic instrument control under a view-limiting device	1.
100-106 Air 92-98	[] Instrument unusual-attitude recognition and recovery	1.
106-116 Air 98-108	[] Return, stabilized approach and final landing	1.
116-120 Ground	[] Clear, taxi in, shut down and secure	1.

Flight = complete movement sequence. Air = minutes from takeoff; ground rows do not advance the airborne clock. Device rows create no airborne time.

Teach lesson 59

Use a complete current-ACS plan of action, the knowledge-test deficiency areas and independent performance evidence. This adaptation is a readiness assessment, not issuance of a pilot certificate or a school graduation authorization.

Source clarification: The 120-minute target may need a continued assessment for the full ACS scope. Instrument time is an 18-minute overlapping subset. Do not issue a practical-test recommendation until the required tasks, knowledge-test deficiencies, experience and endorsements have all been satisfied.

Single-engine airplane, Part 61 adaptation. Confirm speeds, power, configuration, maneuver limitations, recovery altitudes and runway performance from the actual POH/AFM and installed supplements with the CFI before flight. The times are planning allocations; safe operation and demonstrated proficiency govern continuation.

Objectives

- Use a complete current-ACS plan of action, the knowledge-test deficiency areas and independent performance evidence. This adaptation is a readiness assessment, not issuance of a pilot certificate or a school graduation authorization.
- Show independent preparation, control, hazard recognition and safe decisions; record safety interventions and tasks not observed.

Before the flight

- Complete prerequisite ground preparation and demonstrate safe control before adding the next maneuver.
- Use the previous flight's unresolved items to decide whether this mission is ready to begin.
- Finish required training and prepare an assessor plan of action. Identify prior deficiencies and unobserved tasks before the assessment.
- Aircraft instruments are serviceable; a qualified instructor provides lookout/control supervision for view-limited VFR training.

Parked preflight • 15 minutes

- Student checks actual aircraft documents/maintenance status, weather/NOTAMS, loading and fuel; instructor verifies suitability before approving movement.
- Use the POH walkaround and fuel/oil checks. Assess defects instead of assuming that a checklist tick means airworthy.
- Set seats/restraints, controls, radios/charts and loose items; give an aircraft-specific passenger/control-transfer briefing. Stage-appropriate independence is assessed here.

Briefing • 80 minutes

- Use the actual aircraft POH/AFM and supplements to agree on speeds, configuration limits, takeoff/landing margins and safe recovery boundaries. Confirm weather, airspace, fuel, personal readiness and a return/diversion plan.

- Agree on positive control transfer, who maintains the traffic scan and which simulations are permitted. No intentional spins, surprise engine shutdowns or unsafe low-altitude emergency simulations.
- Preview today's ordered flight blocks: Navigate by landmarks and magnetic compass; Choose and fly a diversion; Recover an uncertain position; Steep turns in both directions; Rectangular course: wind correction by leg; S-turns across a road; Turns around a point; Slow flight: configuration, coordination and exit; Full power-off stalls: approach scenario; Full power-on stalls: departure scenario; Basic instrument control under a view-limiting device; Instrument unusual-attitude recognition and recovery.
- Explain angle of attack, warning/full-stall cues, coordination and the conditions that lead from a stall to a spin. Discuss the aircraft-approved recovery; this flight demonstrates stall prevention/recovery, not intentional spins.
- State the assessment rules before starting: no coaching during the attempt, safety intervention is documented, and unobserved/unsafe tasks require more training or a continued assessment. Use current ACS standards where applicable; Stage I uses the solo-readiness gate.
- Use the parked oral to assess certificates/endorsements, airworthiness, current weather, aircraft performance/systems, airspace and human factors. Include airport signs/markings/lighting and emergency/survival equipment where in scope. Final assessments also cover the planned cross-country, night preparation and knowledge-test deficiency codes.
- Brief view-limiter control transfers and instructor lookout. Instrument minutes are inside the flight target, not additional time. Count actual time flown solely by reference to instruments and restore visual flight immediately when safety requires it.

Start, taxi, run-up and runway confirmation · 8 minutes

Teach / coach: Observe student-led surface operations without prompting; safety intervention means further training is needed. Use the actual start checklist, a clear propeller area, immediate engine indication check, early brake check, airport diagram, hold-short discipline and the before-takeoff checklist. Rehearse any ambiguous taxi clearance while stopped.

Student exercise: 1. Start using the POH checklist and verify indications. 2. Test brakes at an appropriate safe speed. 3. Identify the route and every runway crossing. 4. Use required radio calls/readbacks and obtain crossing clearances. 5. Perform run-up/checks in a suitable location. 6. Confirm runway, wind, takeoff plan and rejection triggers before entry.

Practice: One complete start-to-runway sequence; stop for any uncertain clearance or configuration.

Look for: Correct checklist, route, clearances and runway identified; aircraft remains controlled and no hold line is crossed without authority.

Watch / correct: Heads-down taxi work - stop in a safe place before consulting a chart or programming equipment.

Watch / correct: Run-up hurried because of traffic - wait and complete the checks before taking the runway.

Stop / transition: Do not depart with an unresolved aircraft discrepancy, uncertain clearance, failed brake/engine check or weather/endorsement exceedance.

Reference: POH: normal procedures; AIM: taxi/runway operations; FAA runway-incursion guidance

Takeoff, climb and safe transit · 8 minutes

Teach / coach: Observe an independent departure and traffic scan; give only task/route instructions. Confirm wake separation, runway alignment, aircraft-approved rotation/climb/configuration, crosswind correction and the published/assigned departure. For local work select a legal practice area, clear it and brief recovery boundaries before any maneuver.

Student exercise: 1. Verify traffic, clearance when required and the correct runway. 2. Apply the approved takeoff technique while maintaining centerline and wind correction. 3. Climb at the POH/briefed condition and change configuration as approved. 4. Follow the published/assigned departure or pattern. 5. Level off, trim, complete checks and keep collision-avoidance scans.

Practice: One normal departure; use crosswind technique only within the aircraft and lesson authorization.

Look for: Maintains runway/ground track, a safe climb and separation, then arrives at the intended area with the airplane configured and trimmed.

Watch / correct: Letting the nose drift on power application - use coordinated directional control and stop/reject if required.

Watch / correct: Task briefing during a critical climb - finish the climb/checks before discussing the practice exercise.

Stop / transition: Reject for a real takeoff abnormality per the briefed POH plan; discontinue training for wake, weather, traffic, fuel or terrain/airspace risk. No unbriefed low-altitude engine-failure simulation.

Reference: AFH: normal/crosswind takeoff; POH: climb; AIM: traffic/wake avoidance

Navigate by landmarks and magnetic compass · 12 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Orient the chart and identify two independent position clues. 2. Fly the planned heading with wind correction. 3. Mark actual checkpoint time and revise ETA/fuel. 4. Scan weather and airspace ahead. 5. Explain the next landmark and landing alternative; make the assigned radio contact at a low-workload time.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains a continuous position estimate, recognizes drift early and updates ETA/fuel while keeping an outside scan.

Watch / correct: Mistaking one similar feature for the checkpoint - verify with a second independent landmark.

Watch / correct: Ignoring time/fuel after a wind change - update groundspeed, ETA and reserves at the next checkpoint.

Stop / transition: Divert or terminate the exercise for deteriorating weather, uncertain position, unsafe fuel margin or airspace risk; GPS may be used by the instructor for safety without replacing the assigned visual-navigation exercise.

Reference: PHAK: navigation; AFH: cross-country flight; current charts, Chart Supplement and AIM

Choose and fly a diversion · 10 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. State the reason to abandon the original plan. 2. Choose a reachable suitable airport. 3. Estimate heading, distance, time and fuel. 4. Check terrain, airspace, weather and runway information. 5. Fly toward it, refine the estimate and explain the new decision gates.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Chooses early, keeps flying while planning and selects an alternative with safe fuel, terrain and weather margins.

Watch / correct: Choosing the nearest symbol without checking suitability - verify runway, weather and airport status.

Watch / correct: Continuing the old course while calculating - establish a safe interim heading and workload plan.

Stop / transition: End the scenario and provide needed assistance if actual position, weather, terrain or fuel margin is threatened. Do not continue toward a fictitious hazard.

Reference: PHAK: navigation and aeronautical decision-making; Private ACS: Diversion

Recover an uncertain position · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Admit uncertainty promptly. 2. Maintain control and a safe altitude/course. 3. Check last known position, elapsed time and fuel. 4. Use appropriate navigation/ATC assistance to establish position. 5. Choose a safe landing plan rather than wandering.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes loss of confidence, asks for help early and re-establishes position without compromising control, fuel or airspace.

Watch / correct: Continuing random turns while searching - stabilize the airplane and work from the last known fix.

Watch / correct: Delaying a radio request through embarrassment - state uncertainty and request assistance early.

Stop / transition: Stop any simulation before actual uncertainty occurs; for a real lost condition, prioritize safety and obtain assistance. Never intentionally get lost on solo.

Reference: PHAK: lost procedures; AIM: assistance and communications

Steep turns in both directions · 8 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and choose altitude, airspeed and rollout reference. 2. Establish the briefed bank with coordinated controls. 3. Maintain the outside pitch picture and cross-check altitude. 4. Complete the turn, roll out and stabilize. 5. Repeat in the other direction after clearing.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Controls bank, pitch and power without fixation; introductory performance may need coaching, while final assessment uses current ACS tolerances.

Watch / correct: Trying to correct altitude with bank alone - restore the briefed bank and make a small pitch/power correction.

Watch / correct: Looking only inside - return to the outside pitch reference and make short instrument cross-checks.

Stop / transition: Roll out if bank, load, airspeed or altitude approaches an aircraft limit or safe practice boundary; never chase a tolerance through unsafe control inputs.

Reference: AFH: steep turns; Private ACS: Steep Turns

Rectangular course: wind correction by leg · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and establish the briefed entry. 2. Hold parallel ground tracks with wind correction. 3. Change bank smoothly through each corner to preserve spacing. 4. Call the wind correction on each leg. 5. Exit without descending or tightening toward obstacles.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains safe altitude and reasonably uniform spacing, predicts drift and changes bank smoothly rather than skidding the turn.

Watch / correct: Turning too late on the fast downwind corner - anticipate the corner and use an appropriate coordinated bank.

Watch / correct: Looking continuously at the field - alternate the reference with traffic, horizon and altitude scans.

Stop / transition: Exit toward a safe area for traffic, inadequate obstacle clearance, excessive bank or an unstable altitude trend.

Reference: AFH: rectangular course; Private ACS: Ground Reference Maneuvers

S-turns across a road · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and approach the reference as briefed. 2. Begin a coordinated turn that compensates for the wind. 3. Cross wings level and reverse the turn. 4. Adjust bank to create the second comparable arc. 5. Exit and describe the strongest drift correction.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Crosses the reference with wings approximately level, varies bank with wind and preserves altitude and coordination.

Watch / correct: Holding one bank all the way around - anticipate changing groundspeed and vary bank.

Watch / correct: Rushing the reversal with rudder - level the wings at the reference, then enter the opposite bank together.

Stop / transition: Exit if bank/altitude trend is unsafe or the chosen road has traffic/obstructions that compromise the practice area; maintain legal clearances.

Reference: AFH: S-turns; Private ACS: Ground Reference Maneuvers

Turns around a point · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and choose a safe radius and altitude. 2. Enter as briefed. 3. Adjust bank to maintain the ground radius. 4. Keep the ball centered and scan outside. 5. Exit after a full circuit and predict corrections for a second direction.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Holds a consistent radius using coordinated bank changes, with safe altitude and a sustained traffic scan.

Watch / correct: Tightening toward the point when blown inward - reduce bank and re-establish the planned radius.

Watch / correct: Fixating on the point - scan the horizon and traffic, then check the ground reference briefly.

Stop / transition: Leave the maneuver if the required bank exceeds the practice limit, an escape route is compromised or traffic/terrain clearance becomes uncertain.

Reference: AFH: turns around a point; Private ACS: Ground Reference Maneuvers

Slow flight: configuration, coordination and exit · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and confirm the recovery altitude. 2. Set the briefed configuration and slow-flight airspeed. 3. Trim and hold heading/altitude. 4. Make gentle coordinated turns both ways. 5. Recover to cruise using the POH sequence; explain the increasing angle of attack and control pressures.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes reduced control effectiveness, stays coordinated and exits promptly when asked; use the current ACS condition for advanced assessment, not obsolete continuous-warning technique.

Watch / correct: Bank increasing while the ball moves off center - reduce bank and coordinate rudder before continuing.

Watch / correct: Retracting flaps abruptly - use the approved sequence and manage the resulting pitch/sink change.

Stop / transition: Recover immediately for unexpected stall warning, a loss of coordination, an unsafe altitude trend or the briefed recovery floor. No intentional spin entry.

Reference: AFH: slow flight; Private ACS: Slow Flight

Full power-off stalls: approach scenario · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and brief the recovery floor. 2. Select the assigned approved flap configuration. 3. Establish the approach-type descent and enter the stall with coordinated controls. 4. Recognize the full stall, recover promptly and return to climb/cruise. 5. Repeat the alternate configuration only after a stable reset.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the full stall, promptly unloads and restores controlled flight without a secondary stall; assistance level follows the lesson gate.

Watch / correct: Delaying the unload while reaching for controls - reduce angle of attack before attending to configuration.

Watch / correct: Retracting all flaps while still sinking - use the approved incremental recovery sequence.

Stop / transition: Terminate for yaw, a wing drop that exceeds the briefed demonstration, an aircraft limitation or the recovery floor; no intentional spin demonstration.

Reference: AFH: power-off stalls; Private ACS: Power-Off Stalls; POH

Full power-on stalls: departure scenario · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Clear and select the approved entry configuration and power. 2. Establish a coordinated departure-type entry. 3. Identify the full-stall indication. 4. Reduce angle of attack, maintain directional control and recover. 5. Reset before any approved turning entry.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Maintains coordination through entry and recovery, recognizes the stall and avoids a secondary stall while returning to a safe climb.

Watch / correct: Rudder pressure unchanged as speed decreases - actively cross-check coordination and adjust rudder.

Watch / correct: Aiming for an extreme pitch attitude - use the approved entry and judge the stall by aerodynamic cues.

Stop / transition: Recover immediately for developing yaw/roll divergence or the recovery floor; instructor takes control before a spin tendency develops. No intentional spins.

Reference: AFH: power-on stalls; Private ACS: Power-On Stalls; POH

Basic instrument control under a view-limiting device · 12 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Accept the view limiter only after control/communication agreement. 2. Stabilize straight-and-level flight using the instruments. 3. Fly assigned heading changes. 4. Climb and descend at the assigned airspeed and level off. 5. Recover normal visual flight when instructed.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Controls the airplane by reference to instruments with the assistance appropriate to this lesson, uses small corrections and transfers control positively.

Watch / correct: Fixating on one instrument - cross-check attitude, performance and supporting instruments.

Watch / correct: Making large alternating corrections - return to the known attitude/power and make a small correction.

Stop / transition: Remove the view limiter or transfer control immediately for traffic, turbulence, spatial disorientation or unsafe control; instructor retains outside vigilance. No actual IMC without appropriate qualifications/clearance.

Reference: AFH: basic instrument control; Private ACS: Basic Instrument Maneuvers

Instrument unusual-attitude recognition and recovery · 6 minutes

Teach / coach: Assign this task under the prebriefed plan of action, retain safety lookout and observe independent execution. Look for this task-specific result: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits. Apply the current ACS or Stage I solo-readiness criteria as appropriate. Do not demonstrate, hint or coach; document any safety intervention and mark unobserved elements for a continued assessment.

Student exercise: 1. Identify nose-high or nose-low from the instrument cross-check. 2. Apply the appropriate power, pitch and bank recovery sequence. 3. Avoid excess load or airspeed. 4. Re-establish stable flight. 5. Explain the cue that made the attitude clear.

Practice: One independently performed task sequence; repeat only under the assessor's stated policy. A coached retry is training, not the original independent result.

Look for: Recognizes the attitude correctly and restores controlled flight without pulling through excessive bank or exceeding aircraft limits.

Watch / correct: Treating a nose-low/high-bank condition by pulling first - use the correct bank/pitch recovery sequence.

Watch / correct: Using a single instrument without cross-check - verify the attitude with the supporting indications.

Stop / transition: Instructor takes control immediately for incorrect recovery, excessive speed/load, disorientation or an approaching safe-altitude boundary; no extreme attitudes or intentional spins.

Reference: AFH: unusual attitudes; Private ACS: Recovery from Unusual Flight Attitudes; POH

Return, stabilized approach and final landing · 10 minutes

Teach / coach: Assess an independent arrival and landing; provide safety intervention if needed and record it. Recheck weather/traffic, airport/runway identity, landing performance, pattern entry and the aircraft-specific stabilized-approach gate. Teach an early go-around decision whenever safe landing is uncertain.

Student exercise: 1. Secure maneuver/route work and complete descent checks. 2. Confirm airport/runway, wind and pattern/clearance. 3. Join with traffic awareness and an appropriate radio report. 4. Fly the approved stable approach and manage wind correction. 5. Land aligned and maintain control through rollout, or go around early.

Practice: One complete arrival and final landing; include a go-around whenever the real approach fails the briefed gate.

Look for: Maintains traffic separation and a stable approach; touches down aligned with adequate runway margin or makes an early safe go-around.

Watch / correct: Forcing the landing to finish on time - go around and accept additional flight time.

Watch / correct: Fixating on the aiming point - maintain airspeed, alignment, traffic and runway margin in the scan.

Stop / transition: Go around for unstable path/speed, drift, clearance uncertainty or unsafe runway margin; divert if weather, fuel or lighting makes return unsuitable.

Reference: AFH: traffic patterns and landings; POH; AIM: airport operations

Clear, taxi in, shut down and secure · 4 minutes

Teach / coach: Observe after-landing and securing without distraction; inspect the learner's checklist completion and aircraft condition. Keep the airplane controlled until clear of the runway, then use after-landing/taxi/shutdown checklists. Account for keys, tie-downs/chocks, discrepancies and postflight reporting.

Student exercise: 1. Clear the runway fully and stop when needed before after-landing checks. 2. Follow the cleared/briefed taxi route with wind correction. 3. Park and shut down using the POH checklist. 4. Secure the aircraft and inspect for discrepancies. 5. Report actual tasks/counts and any concern before leaving.

Practice: One complete runway-exit-to-secured-aircraft sequence; parked debrief follows this flight block.

Look for: Runway is clear before heads-down work; aircraft is safely shut down/secured and discrepancies are communicated.

Watch / correct: Doing cleanup during rollout - maintain control and wait until clear.

Watch / correct: Leaving a tie-down/checklist item unverified - walk around and confirm the secured condition.

Stop / transition: Stop taxi and request assistance for an uncertain clearance, brake/control problem or obstruction; do not leave an unsecured or unreported discrepant aircraft.

Reference: POH: after landing, shutdown and securing; AIM: taxi

Completion criteria

- All required tasks are observed or explicitly finished in continued training; safe preparation, traffic awareness and control take priority over the nominal time.
- The instructor evaluates the actual student performance and records any assistance or remaining deficiency before advancing the lesson.
- Demonstrate applicable current Private ACS knowledge, risk management and skill without coaching. A missing task or safety intervention prevents a complete assessment; this guide grants no certificate or school examining authority.
- Confirm experience, tests, endorsements and the required preparation within the applicable regulatory period separately from this mock/readiness result. Review and correct knowledge-test deficiency areas before recommendation.

Debrief · 40 minutes

- Ask the student to reconstruct one control/decision sequence and identify the earliest cue of the largest error.
- Compare the first and last attempt using observable aircraft control and decisions; separate assisted practice from independent evidence.
- Record actual flight/instrument time, route and full-stop counts as applicable outside this app. Agree on the next specific exercise and any carry-forward gate.

Repeat or carry forward

- Short-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.

- Soft-field takeoff and landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Forward slip to a stabilized landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Go-around from a rejected landing: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Emergency descent: controlled escape from an urgent hazard: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- Simulated engine failure and emergency approach: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- VOR navigation and radar/ATC services when available: not allocated adequate practice in this target flight. Add a safe dedicated block or another flight before declaring the source lesson complete.
- If a required task is unobserved, unsafe or below the lesson gate, extend/continue training or assessment. Complete and verify it before progressing; elapsed minutes never substitute for proficiency.

Study and sources

Current Private Pilot Airplane ACS: knowledge, risk and skill review

Sporty's Academy Private TCO V20: lesson 59 - source page 84, 85 - Sporty's Academy Private 141 TCO V20 / 06-17-2026